The State of Information Quality and Data Governance
2016 Industry Survey & Report

Understanding Current Trends in the Ways Organizations Manage the Quality of their Information Assets and Compensate IQ Professionals

By

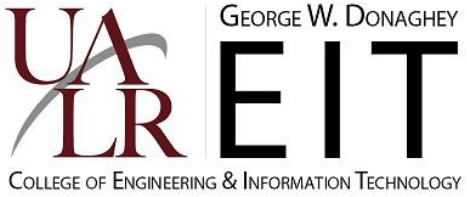
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IQ International, the International Association for Information and Data Quality (IAIDQ)

University of Arkansas at Little Rock, Information Quality Program (UALR-IQ)

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About IQ International, the International Association for Information and Data Quality (IAIDQ)

Chartered in January 2004 the International Association for Information and Data Quality (IAIDQ), now known as IQ International, is a not-for profit, vendor-neutral professional society, and the only international professional organization for information and data quality practitioners. IQ International is advancing the quality of information and data around the world by building a community that supports learning and sharing knowledge for the benefit of all information consumers. IQ International offers a wide range of products and services to meet the needs of information/data quality professionals at all experience levels. To learn more about IQ International, please visit the website: iaidq.org.

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About the Information Quality Program at UALR

The world's first graduate degree program devoted to Information Quality was created by the University of Arkansas at Little Rock in 2006. The program was designed in collaboration with the MIT Information Quality Program directed by Dr. Richard Wang, and a significant portion of the start-up costs were underwritten by Acxiom Corporation under then CEO Charles Morgan. The Information Quality Program at UALR (UALR-IQ) is housed in the Department of Information Science in the Donaghey College of Engineering and Information Technology. The purpose of the UALR-IQ Program is to meet the growing demands from academia, industry, and government for qualified professionals with graduate-level degrees (Certificate, Master's and PhD) that encompass the theory, principles, and practices essential to advancing the discipline of Information Quality and closely-related areas such as entity and identity resolution, identity management, information integration, governance, risk assessment, compliance, and enterprise architecture. To learn more about the IQ Program at UALR, please check out their website: http://ualr.edu/informationquality.

About the IQ International Industry Report Series

This report is part of a series of industry reports jointly produced by IQ International and the Information Quality Program at UALR (UALR-IQ). Earlier reports in this series include the following:

- The State of Information and Data Quality (November 2012) co-authored by Elizabeth Pierce, C. Lwanga Yonke, Piyush Malik, and Chitra Kagathur Nagaraj. Available at http://iaidq.org/publications/pierce-2012-11.shtml.
- The Job of the Information/Data Quality Professional (February 2011) co-authored by C. Lwanga Yonke, Christian Walenta, and John R. Talburt. Available at http://iaidq.org/publications/yonke-2011-02.shtml.
- Information/Data Quality Salary and Job Satisfaction Report (July 2009) co-authored by Elizabeth Pierce, C. Lwanga Yonke, and Apolonia Lintag. Available at http://iaidq.org/publications/pierce-2009-07.shtml.
- The State of Information and Data Governance: Understanding How Organizations Govern their Information and Data Assets (April 2008) co-authored by Elizabeth Pierce, Wendell Scott Dismute, and C. Lwanga Yonke. Available at http://iaidq.org/publications/pierce-2008-04.shtml.

Acknowledgements

The authors and our IQ International research partners would like to thank all the people who contributed to the 2016 Trends in Data Quality and Data Governance Industry Report. We greatly appreciate and thank the many individuals who completed our survey.

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Executive Summary / Key Findings

This report details the findings of a survey conducted by a team of researchers from the UALR-IQ Program and IQ International during early spring 2016. The purpose of the study is to better understand current trends in how organizations are governing their data, managing the quality of their information, and compensating the individuals who perform these services. The results presented in this report are based on the responses of 160 individuals who completed the survey. These individuals represent a wide range of staffing roles, functional areas, organizations, industries, and geographic regions.

Trends in Data Governance

- Over half of the respondents (51.0%) reported their customer data is the best managed in their organizations, followed by Financials (47.7%), Products and Production (33.8%), Employees (33.1%), and Services (27.2%).
- Data governance activities are becoming the norm in most organizations. 40% of survey participants reported their organization had implemented data governance efforts over two years ago. Nearly 32% of respondents said their organizations had implemented data governance within the last two years or were currently implementing data governance efforts. Most survey participants expected their organization's data governance efforts to increase in the coming year.
- For those organizations with data governance councils, survey respondents indicated that there is growing representation from the lines of business, particularly at the senior executive level.
- When it comes to data stewardship, organizations use several different approaches. According to our survey responses, organizing data stewards by lines of business is the most common approach (41.9%), followed by organizing data stewards by data domains (36.3%), and organizing data stewards by IT Systems (26.3%).

Trends in the Relationship Between Data Governance and Data Quality

- Nearly half of survey participants (45.5%) said in their organization data quality activities were viewed as a subset of their organization's data governance efforts.
- "Reducing risk and assuring compliance" was chosen by 46.3% of survey respondents as the main focus for their organization's data governance and data quality efforts. "Increasing Revenue" came in second (20.0%) followed by "Reducing Costs" (15.6%).
- Based on our survey responses, the top five categories of data tools most used by organizations are Data Quality Monitoring Tools (58.5%), Business Glossary/Data Dictionary Tools (55.3%), Data Profiling and Quality Assessment Tools (49.1%), Extract-Transform-Load/Integration Tools (46.5%), and Data Modeling Tools (45.3%).

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Trends in Data Quality

- Data quality is making steady progress, but still has a long way to go. Nearly half of survey participants rated the quality of the data in their organization as Fair (46.2%) with another third (34%) rating their data as good. Only 7.7% of survey participants assessed their organization's data quality as excellent. However, the majority of survey participants (68.6%) stated data quality in their organization has improved either significantly or very significantly over the last two years.
- Respondents reported that they are performing a wide range of information quality tasks, but on a less frequent basis than reported in our 2009 Survey Report with many data quality activities being done on a monthly, quarterly, or annual basis rather than on a daily or weekly basis.
- 61.5% of our survey participants stated that they spend 50% or more of their time each week on activities associated with these information quality domains. These individuals are the focus of the final section of our report – Compensation and Job Satisfaction of Data Quality Professionals.

Compensation and Job Satisfaction of Data Quality Professionals

- Women made up about one third of our survey population of individuals who spend 50% or more of their weekly effort on data quality activities.
- Just under 40% of our survey population of data quality professionals have held their jobs for less than 2 years while a third (33.3%) have held their positions 2 to 4 years, and roughly a quarter (26.9%) had held their positions longer than 4 years.
- Over half of our survey population of data quality professionals (54.3%) are planning to stay with their current positions for now. 60.3% feel either secure or very secure about their present position.
- While Managers and Executives/Owners have seen a substantial increase in their salaries over the last seven years, most Staff/Individual Contributors and Team Leaders/Supervisors have seen the salaries for their positions decrease over the last several years.
- The majority of our survey population of data quality professionals are either satisfied or very satisfied with their compensation (60.8%), and are either satisfied or very satisfied with their job responsibilities (78.5%).
- Over 82% of our survey population of data quality professionals experience some level of significant stress regarding their jobs.
- Nearly 80% of our survey population of data quality professionals anticipates that their data quality activities will increase over the next 12 months.
- The biggest obstacle faced by our survey population of data quality professionals (73.9%) is the challenge posed by how to demonstrate the value of data quality to their employers.

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Introduction

Overview and Objectives

This report presents the findings of a survey jointly conducted by IQ International (IAIDQ) and the Information Quality Program at the University of Arkansas at Little Rock (UALR-IQ) between February 1 and March 15, 2016. The purpose of this year's study is to better understand the current state of data governance and data quality practices in organizations along with the job situation of the professionals who engage in these activities, and how these topics are changing over time. Our goal is to provide valuable insights for information/data quality practitioners, job seekers, employers, and the academic community in evaluating existing conditions, identifying best practices, and setting the agenda for future growth of the information quality discipline. Note: This report uses both "data quality" and "information quality" depending on the context most relevant to a particular situation. In addition, the report uses the label "business" as a synonym for "non-IT/IS".

Survey Methodology

During November and December of 2015, a team of UALR-IQ and IQ International researchers developed a questionnaire to gather information regarding data governance, data quality, and compensation for data quality professionals in today's organizations. The survey was officially launched via SurveyMonkey.com on February 1, 2016. IQ International sent several invitations via e-mail to individuals on its mailing list, asking them to complete the web-based survey. Invitations were also distributed to Alumni of the UALR Graduate Program in Information Quality. In addition, we posted information about the survey on the following data groups on LinkedIn: IQ International Information/Data Quality Professional Open Community, Data Quality and Metadata Management, Data Governance & Data Quality, Data Governance & Stewardship, Enterprise Data Quality, The Data Quality Association, UALR Information Quality Graduate Program, Big Data, Analytics, Hadoop, NOSQL & Cloud Computing Community, Big Data, Analytics, and Business Intelligence & Visualization Experts Community. The survey closed on March 15, 2016.

By the time the data collection period ended, we had tallied 244 survey attempts. Three survey attempts were eliminated based on a high probability that these attempts were duplicates. That is, based on the IP Address, date and time of the attempt, along with a comparison of the responses to the questions, there was strong evidence that these individuals had started the survey, were interrupted, and then returned to complete another survey attempt. An additional 21 survey participants were disqualified from completing the full survey because they indicated they were not currently working full time for any organization. Of the remaining 220 survey responses, 60 participants abandoned the survey before completing any questions related to data governance or data quality in their organization. Since the primary purpose of this report is to focus on the status of data governance and data quality, we decided to focus our attention on the responses of the 160

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individuals who completed the main sections of the survey. Because not every person in this group of 160 individuals answered every question, the number of respondents varies by question. This is specified in the tables and charts featured in this report. In addition for some questions, multiple responses by individuals were permitted. This is designated where applicable in the tables and charts. Finally some survey questions included an “Other” option. The presence of an “Other” option allowed survey participants to write in their own responses in addition to selecting answers from a pre-determined set. Whenever appropriate, this report lists alternative answers provided by respondents when discussing the results for these open-ended questions.

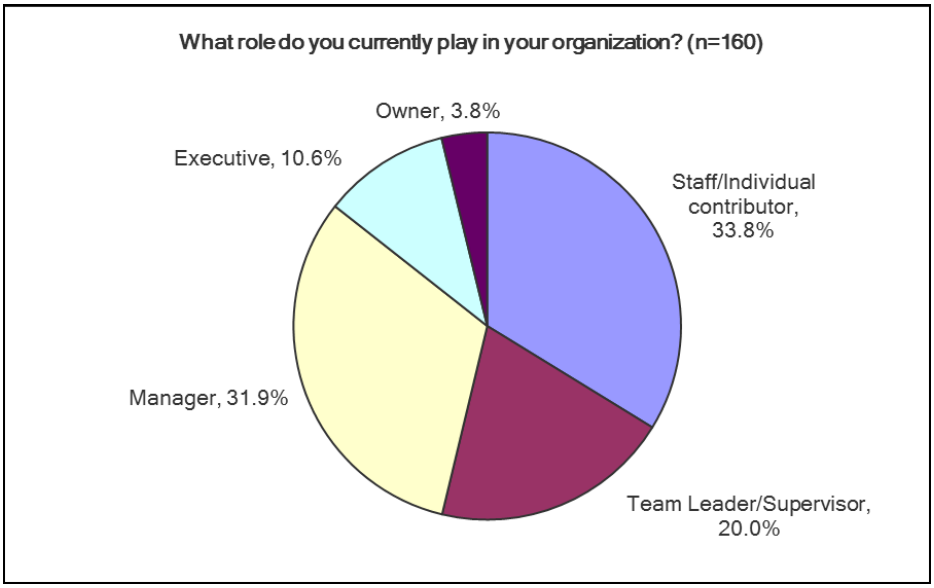
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Survey Demographics

The 160 survey participants who completed our survey represent a diverse set of organizations from around the globe.

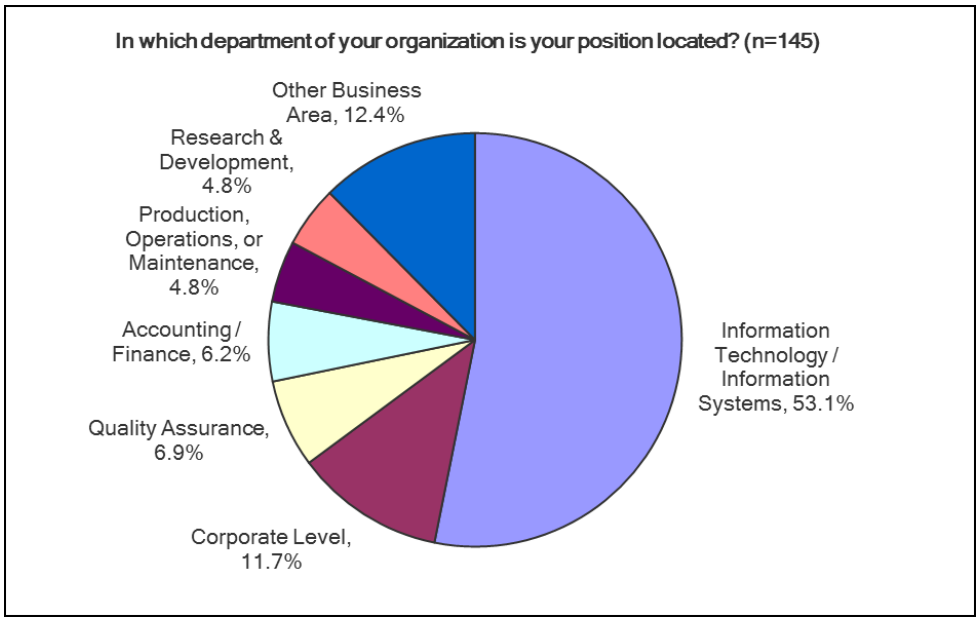
Organizational roles of survey participants: The majority of our survey participants identified themselves as either Staff/Individual Contributor (33.8%), Manager (31.9%), or Team Leader/Supervisor (20.0%). The distribution of the responses for the role of survey participants is consistent with the range of variation seen in past IQ International industry studies for this question.



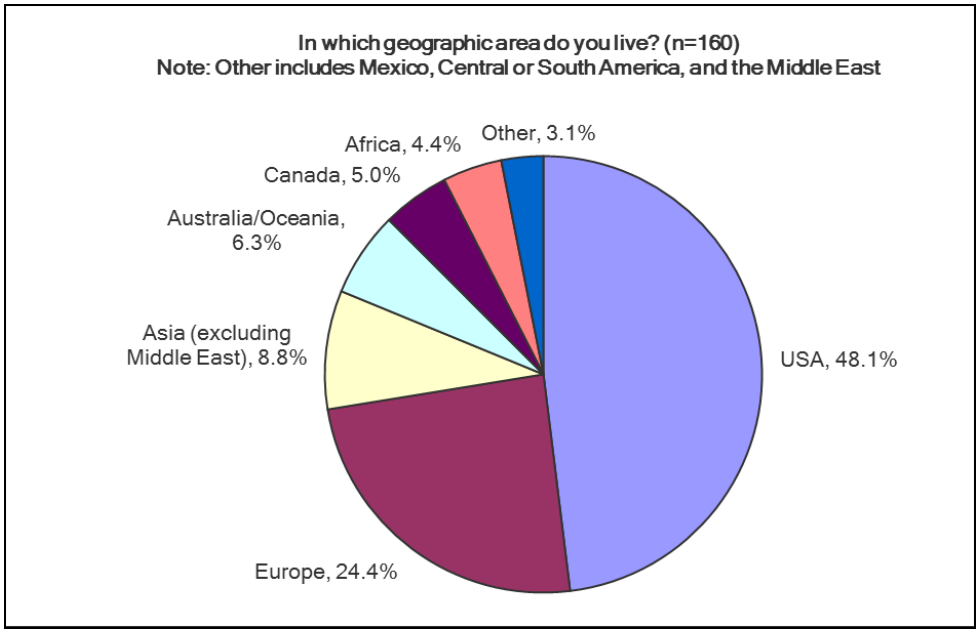
Functional area of survey participants: Most respondents selected either Information Technology/Information Systems (IT/IS) (53.1%), Corporate Level (11.7%), Quality Assurance (6.9%), or Accounting/Finance (6.2%) as the functional area where their position is located. The distribution of the responses for the role of survey participants falls within the range of variation seen in past IAIDQ industry studies for this question with the exception of the IT/IS percentage. This year's IT/IS percentage was higher compared to previous surveys (2008: 42.8%, 2011: 36.7%, and 2012: 37.7%). More investigation is needed to determine whether this difference is due to factors surrounding this year's survey or if it reflects a trend towards greater IT/IS participation in organizations' data governance and data quality efforts.

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Work location of survey participants: The regions with the largest survey participation were the United States (48.1%), Europe (24.4%), Asia (8.8%), Australia/Oceania (6.3%), and Canada (5.0%). The distribution of the responses for the global regions of survey participants is consistent with the range of variation seen in past IAIDQ industry studies for this question. Within the United States, the regions with the most respondents were the South Atlantic States (23.7%), the Pacific States (22.4%), the West South Central States (21.1%) and the Middle Atlantic States (15.8%).



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If you live in the US, please specify the geographic area. (n=76)	
Answer Options	**Response Percent**
South Atlantic (DE, FL, GA, MD, NC, SC, VA, WV)	23.7%
Pacific (AK, CA, HI, OR, WA)	22.4%
West South Central (AR, LA, OK, TX)	21.1%
Middle Atlantic (NJ, NY, PA)	15.8%
West North Central (IA, KS, MN, MO, NE, ND, SD)	5.3%
Mountain (AZ, CO, ID, MT, NV, NM, UT, WY)	5.3%
New England (MA, CT, NH, RI, VT)	3.9%
East North Central (IL, IN, MI, OH, WI)	2.6%
East South Central (AL, KY, MS, TN)	0.0%

Industries associated with survey participants' organizations: Survey participants represented over 25 different industries. The largest participation for this year's survey came from the Banking/Financial Services (15.2%), Energy/Oil & Gas/Utilities (13.9%), Healthcare (9.9%), Consulting/Professional Services (7.9%), Manufacturing (6.6%), Software/Internet (6.6%), Education (6.0%) and Insurance (6.0%) industries. The distribution of the responses for the industry where individuals work falls within the range of variation seen in past IAIDQ industry studies for this question.

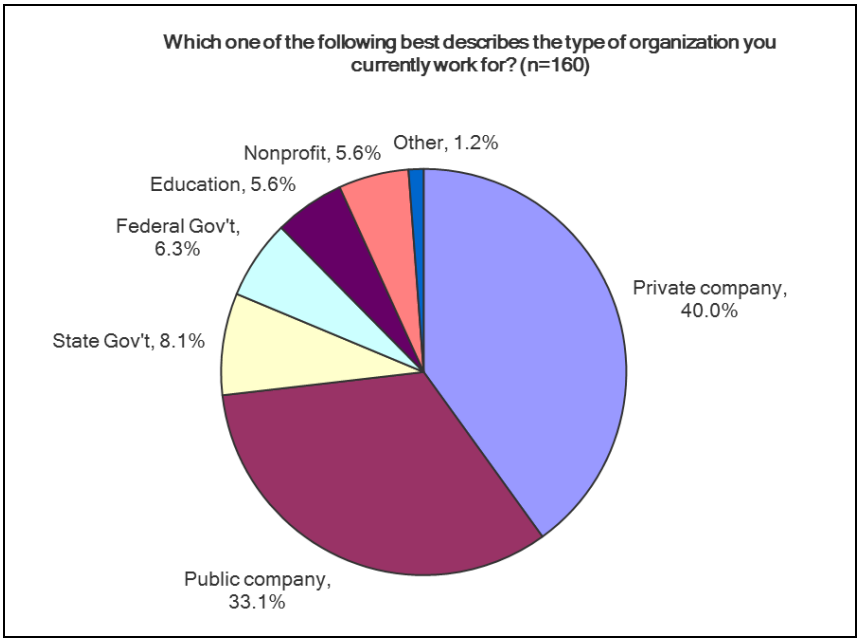
Which of the following best describes your organization's primary industry? (n=151)	
Answer Options	**Response Percent**
Banking/Financial Services	15.2%
Energy/Oil & Gas/Utilities	13.9%
Healthcare	9.9%
Consulting/Professional Services	7.9%
Manufacturing	6.6%
Software/Internet	6.6%
Education	6.0%
Insurance	6.0%
Government: State/Local	4.6%
Government: Federal/National	4.0%
Telecommunications/Communications	4.0%
Retail/Wholesale Distribution	3.3%
Advertising/Marketing/Public Relations	2.0%
Bio-technology/Pharmaceuticals	2.0%
Media/Entertainment/Publishing	2.0%
Construction/Architecture/Engineering/Real Estate	1.3%
Food/Beverage/Agriculture	1.3%
Law	1.3%
Logistics/Transportation	1.3%
Hospitality/Travel	0.7%

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Table Note: 12 Individuals provided other industry descriptions such as Railways, Vehicle Fleet Management, Automotive/Marketing, Healthcare Technology, Analytics, and Security.

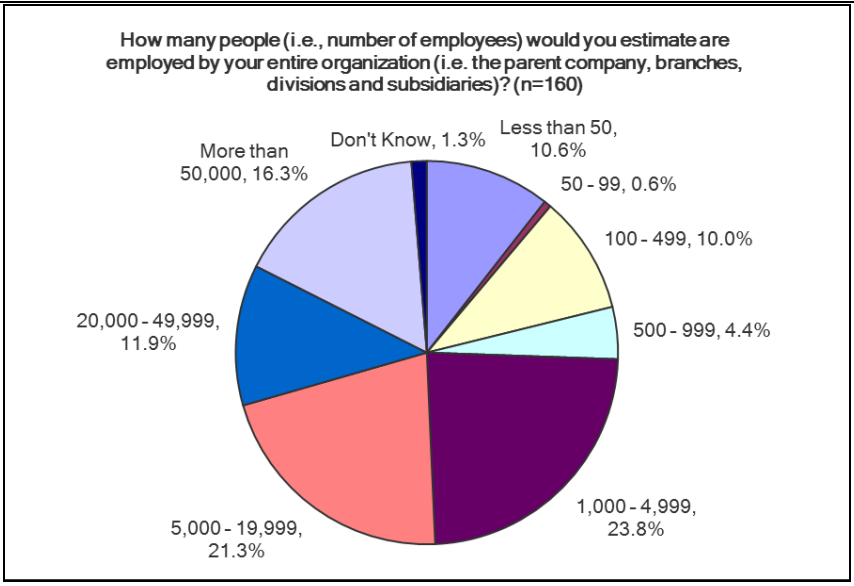
Type of organizations where survey participants work: Most survey participants work for either privately held (40.0%) or publicly traded companies (33.1%). The response distribution for the type of organizations where individuals work is consistent with past IAIDQ industry studies for this question.



Organization size by employee count: Approximately half of the survey participants (49.5%) reported being employed by large organizations (more than 5,000 employees), roughly a quarter (23.8%) reported being employed by medium size organizations (1,000 to 4,999 employees) and the remainder (25.6%) reported being employed by smaller organizations. The response distribution for the organization size is consistent with past IAIDQ industry studies for this question.

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Trends in Data Governance

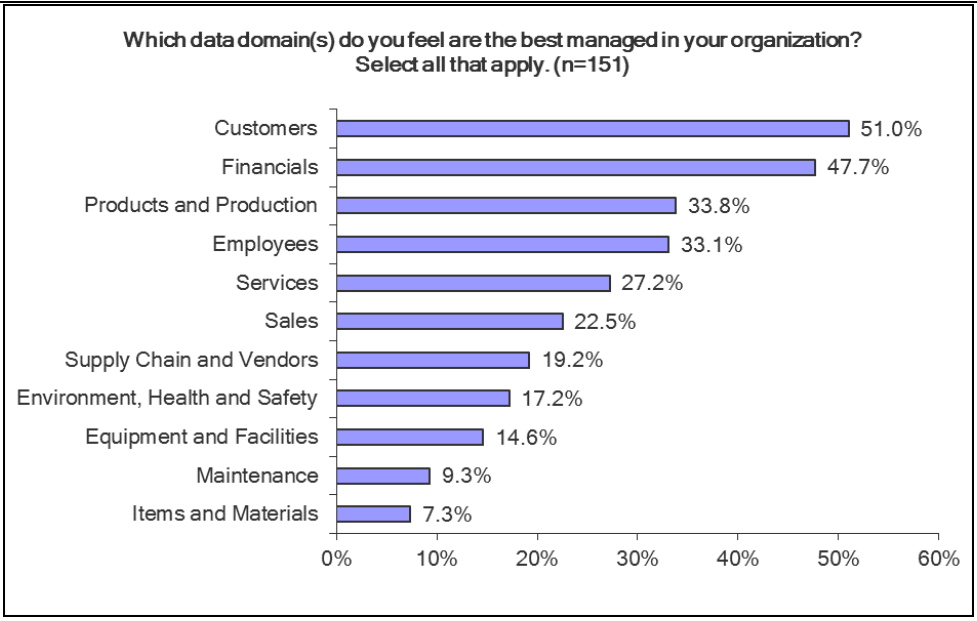
In this section we examine how perceptions toward data governance have evolved over the last several years. Trend charts comparing results from previous surveys are included in the Appendix.

Which data domains are receiving the most attention?

Over half of the respondents (51.0%) reported their customer data is the best managed in their organizations, followed by Financials (47.7%), Products and Production (33.8%), Employees (33.1%), and Services (27.2%). The phrasing of this question differed in this year's survey. Previously we had asked "On which subject areas do your information quality/data quality efforts focus?" Compared to our 2008 and 2012 surveys, the order of the data areas is very similar; however, this year's percentages are not as high. This would seem to suggest that while one would hope that more attention on a subject area would translate into better managed data, that outcome is not guaranteed.

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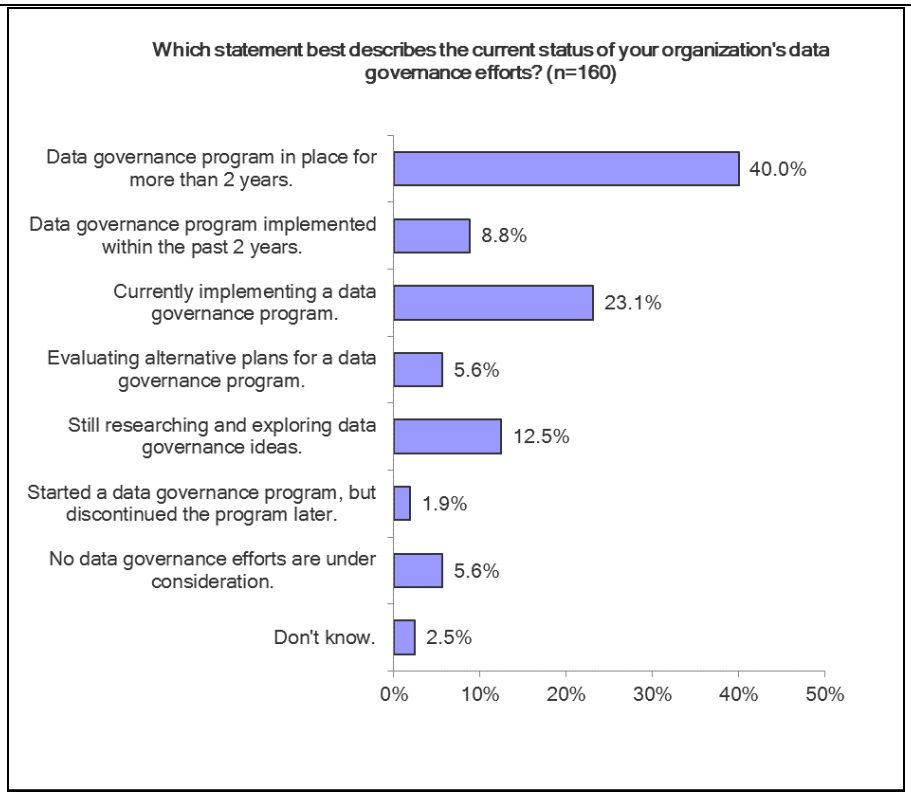


What is the current status of data governance in organizations?

Data governance activities are becoming the norm in most organizations. 40% of survey participants stated their organization had implemented data governance efforts over two years ago. Nearly 32% of respondents said their organizations had implemented data governance within the last two years or were currently implementing data governance efforts. Only a quarter (25.6%) indicated their organization was still exploring data governance initiatives or were not making progress in data governance efforts. This is a big change from our 2008 survey when we asked a similar question. Back then only 8.8% of the survey participants reported their first data governance efforts had been in place for more than 2 years and over half (52.5%) said their organization was still exploring data governance initiatives or were not making progress in their data governance efforts.

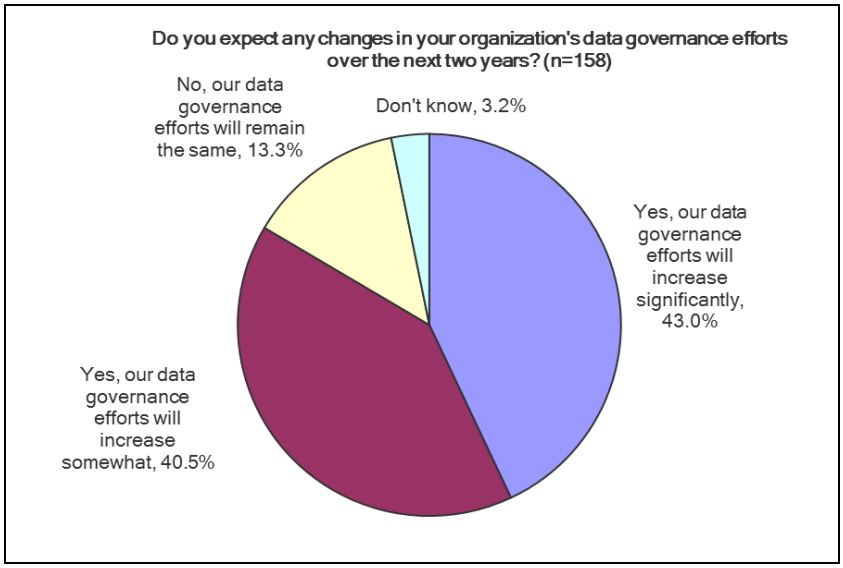
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What changes are expected in data governance efforts over the next two years?

The majority of survey participants (83.5%) expected their organization's data governance efforts to increase (either significantly or to some extent). No one believed their data governance efforts would decrease. A small percentage (13.3%) felt their organization's data governance efforts would remain the same and only 3.2% did not know the direction their organization's data governance efforts would take. The distribution of this year's responses regarding expectations for changes in data governance efforts is very similar to how individuals responded in 2008 when we asked this question.



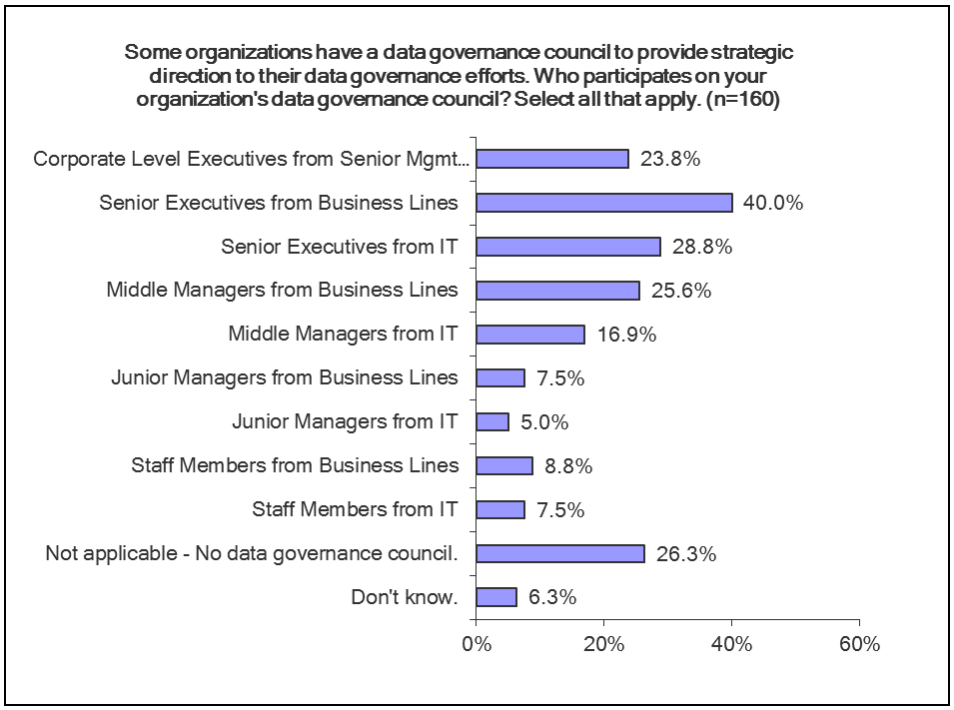
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Who is participating in setting the direction for data governance in organizations?

26.3% of respondents said their organizations did not have any sort of data governance council. Of those whose organizations did have some type of data governance council, survey participants selected personnel from the lines of business with greater frequency (81.9% combined percentage) as serving on their organization's data councils compared to IT personnel at various levels (58.2% combined percentage). Senior executives (68.8% combined percentage) appear to be the most likely members on data governance councils followed by middle managers (42.5% combined percentage) and corporate level executives (23.8% combined percentage). Junior level personnel and staff members appear to be the least likely members of a data governance council.

In both 2004 and 2008 we posed similar versions of this question in our surveys. Over the years we have adjusted the response categories to try to better reflect the various combinations of member participation. For example in this year's 2016 survey we added the categories of Staff and we changed the option of "Senior/Executive Management Team is the top IQ / Data Governance body" to "Corporate Level Executives from the Senior Management Team" to reflect that C-level executives may be serving on data governance councils with a diverse membership. Comparing the responses from 2004, 2008, and 2016 there are several trends to watch. While the majority of categories have remained fairly stable over the years, the percentages for participation by middle managers from business and middle managers from IT have both declined. However the percentage for participation by Senior Executives from the Business Lines is sharply up in 2016. This shift may reflect that organizations are making data governance a higher strategic priority for their executive leaders. This is something that we will need to continue to monitor in future surveys.



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What is the role of data stewards in an organization's data governance efforts?

When it comes to assigning data stewards, respondents indicated that organizing data stewards by lines of business is the most common approach (41.9%), followed by organizing data stewards by data domains (36.3%) or IT Systems (26.3%). A quarter of survey participants (25.0%) reported their organization has a dedicated data governance office or chief data steward. This question is new to our surveys and it is meant to provide insight as to how organizations are assigning their data stewards. Individuals were able to choose more than one option for this question indicating that many organizations are using multiple approaches for allocating their data stewards.

Some organizations have designated data stewards to help manage the data and to ensure its fitness of use. Select any of the following statements which are true about data stewardship roles in your organization. Select all that apply. (n=160)	
Answer Options	**Response Percent**
My organization has data stewards who are associated with various lines of business units (e.g., sales, marketing, human resources).	41.9%
My organization has data stewards who are associated with specific data domain (e.g., customers, employee, product, vendors).	36.3%
My organization has data stewards who are associated with various IT Systems (e.g., ERP, MDM, or Financials).	26.3%
My organization has a dedicated data governance officer (a.k.a. chief data steward) responsible for overall consistency of the data governance / stewardship program across the organization.	25.0%
My organization does not have any data stewards.	16.9%
Don't know.	5.0%

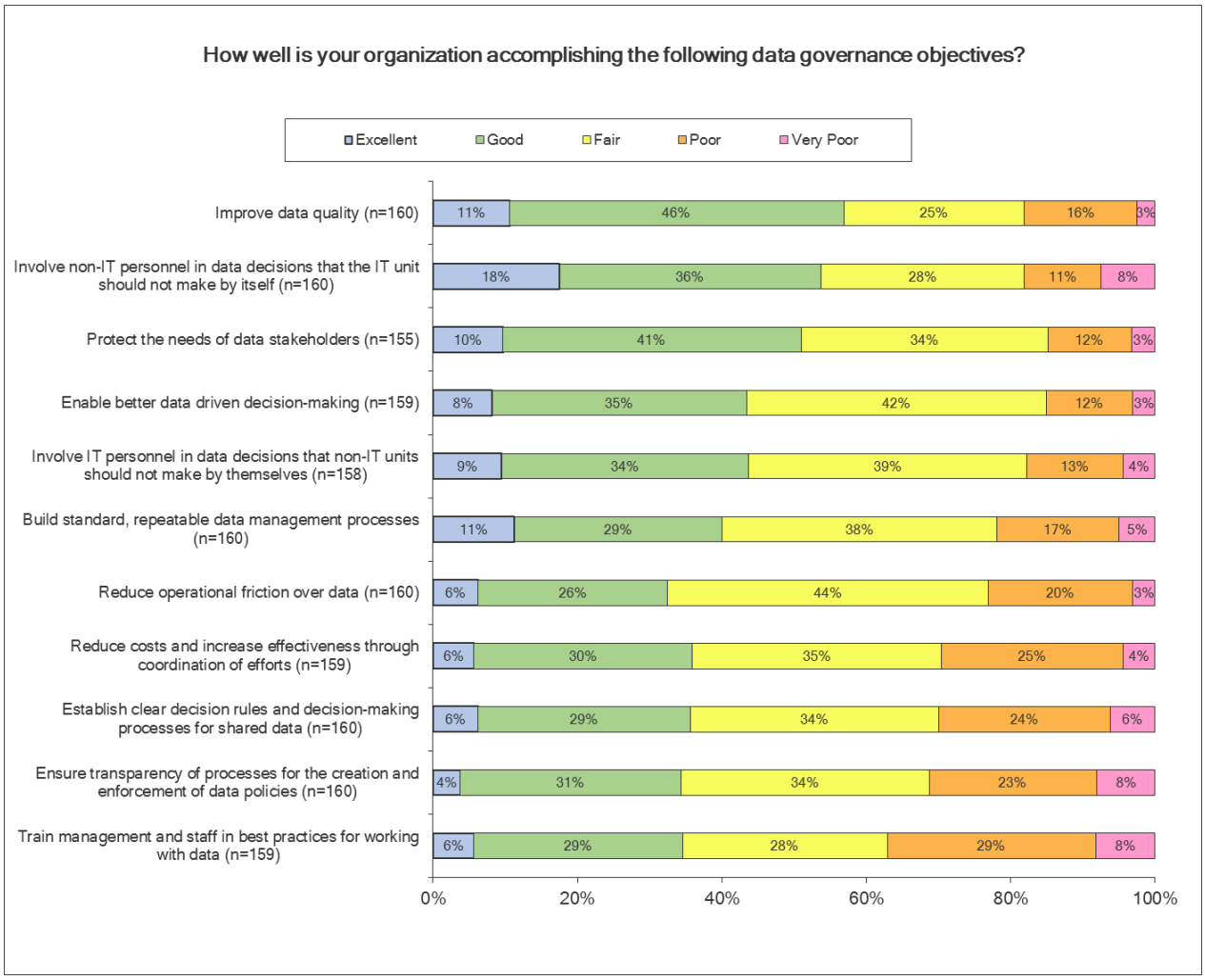
How well are organizations achieving their desired data governance results?

In the 2008 and 2012 surveys we asked participants to rate the effectiveness of their data quality and data governance efforts by considering the amount of goals met. In both surveys about a quarter of respondents rated the effectiveness as good to excellent (most to all goals were being met), about half of the respondents rated the effectiveness as OK (some goals were being met), and the remaining quarter rated the effectiveness as very poor to poor (no to few goals were being met). This year we wanted to get more insight from individuals as to which data governance goals were being achieved. Of all the data governance objectives listed, the three options that received a majority of "good" or "excellent" ratings were "Improve data quality" (~57%), "Involve non-IT personnel in data decisions that the IT unit should not make by itself" (~54%) and "Protect the needs of data stakeholders" (~51%). Most of the other objectives of an organization's data governance efforts were mainly rated as "fair" by

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our survey participants. The data governance objective with the overall lowest rating was “Train management and staff in best practices for working with data” (~37% of respondents rated that objective’s achievement as either “poor” or “very poor”). These results show that the majority of organizations are having difficulties meeting the data governance objectives listed in our survey.



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Relationship between Data Governance and Data Quality

This section of the report explores some current trends regarding the relationship between Data Governance and Data Quality activities in organizations. Trend charts comparing results from previous surveys are included in the Appendix.

What is the relationship between data governance and data quality efforts?

Nearly half of survey participants (45.5%) said in their organization data quality activities were viewed as a subset of their organization's data governance efforts. For another 21.4%, the relationship was reversed (data governance activities were viewed as a subset of their organization's data quality efforts). For 20.8% of respondents, the data quality activities and data governance activities were considered separate but coordinated activities. Only 12.3% reported no relationship between their organization's data quality and data governance efforts.

In earlier surveys we had asked respondents to tell us about the reporting relationship between the people leading the data governance and data quality efforts. In the 2008 and 2012 surveys, the average response distribution divided roughly into three segments. About a third of respondents said that data governance and data quality were led by the same person; about a third of respondents said the efforts were led by different people (who reported to either the same or different managers); and about a third said either there was no specific person in charge or there were other arrangements in place. In this year's survey question we were curious to know more specifically, what is the organizational perception between data governance and data quality activities? It appears for now that in many companies, data governance is considered the driver for data quality activities.

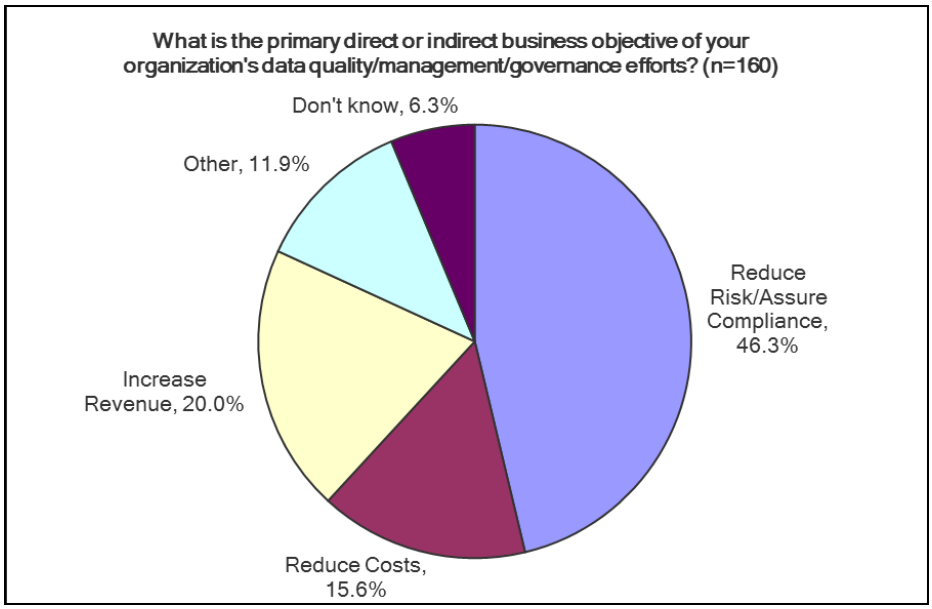
Which statement best reflects the relationship between your organization's Data Quality activities and its Data Governance activities? (n=154)	
Answer Options	**Response Percent**
Our data quality activities are considered a subset of our organization's data governance efforts.	45.5%
Our data governance activities are considered a subset of our organization's data quality efforts.	21.4%
Data quality and data governance activities in our organization are considered separate efforts but there is some level of coordination between the two initiatives.	20.8%
Any data quality or data governance activities in our organization are considered separate efforts and are completely independent of each other.	12.3%

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What is the primary business objective of data governance and data quality efforts?

"Reducing risk and assuring compliance" was selected by 46.3% of survey participants as the main focus of their organization's data governance and data quality efforts. "Increasing Revenue" came in second (20.0%) followed by "Reducing Costs" (15.6%). The distribution of this year's responses for the primary business objective of data governance efforts is fairly consistent with how individuals responded in our 2011 and 2012 surveys, but there is one trend to watch. While reducing risk and assuring compliance remains the most chosen primary objective, this year "Increasing Revenue" edged out "Reducing Costs" as the second most chosen primary objective. Previously in 2011 and 2012, "Reducing Costs" had been the second most common primary objective. This may indicate more organizations are shifting their main data management goal from cost containment to revenue generation.

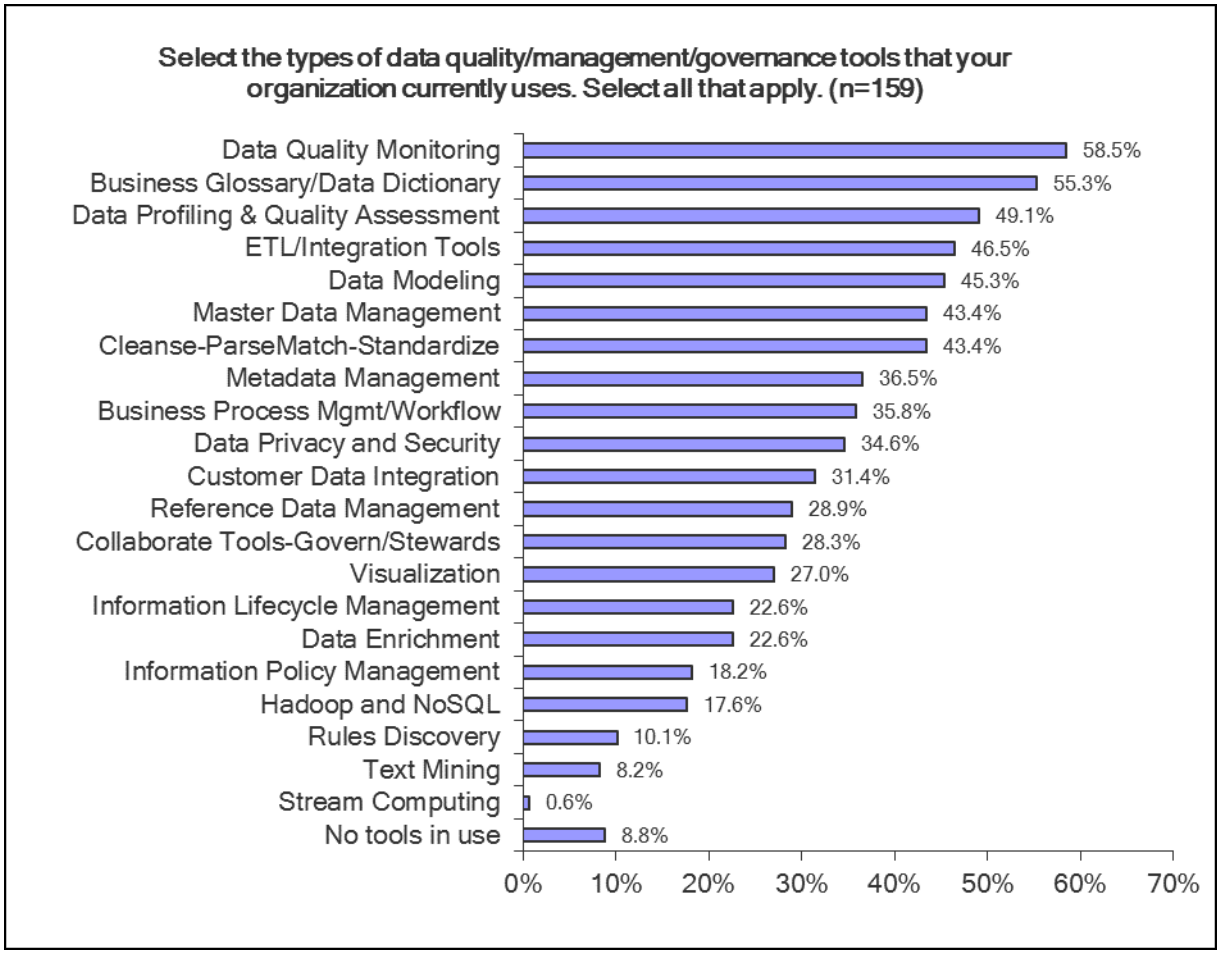


What are the categories of data tools most used by organizations?

According to our survey respondents, the top categories of data tools most used by organizations are Data Quality Monitoring (58.5%), Business Glossary/Data Dictionary (55.3%), Data Profiling and Quality Assessment (49.1%), Extract-Transform-Load/Integration (46.5%), and Data Modeling (45.3%). The relative rankings of the most frequently chosen category of data tools have been fairly consistent since we started asking this question on our 2008 survey. Over the years we have adjusted the names of the categories to keep pace with changes in the data tools marketplace. For example this year we added the category "Business Glossary/Data Dictionary" which ended up in this survey's top five list of data tools.

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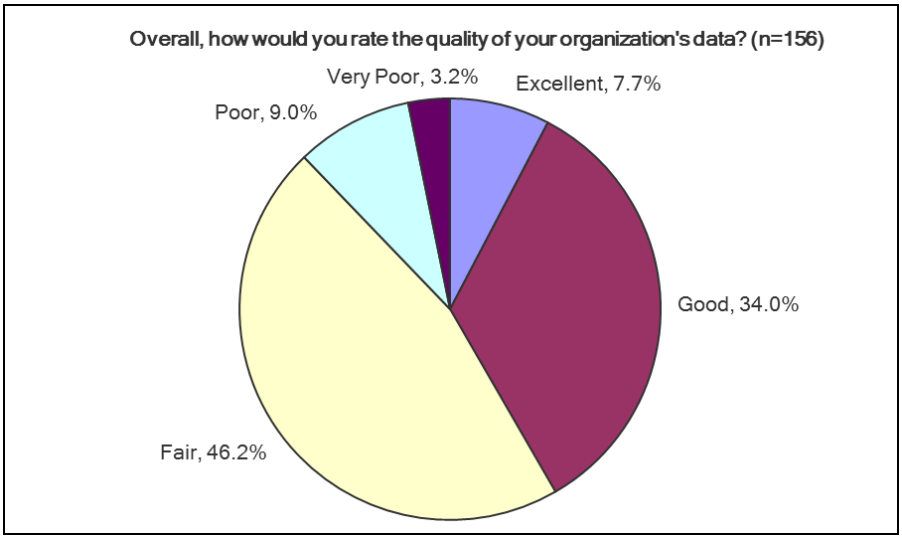
Trends in Data Quality

This section of the report describes the attitudes towards data quality in today's organizations. Trend charts comparing results from previous surveys are included in the Appendix.

How good is the data?

Data quality still has a long way to go. Nearly half of respondents rated the quality of the data in their organization as Fair (46.2%) with another third (34%) rating their data as good. Only 7.7% of survey participants assessed their organization's data quality as excellent. As a comparison, in 2008 and 2011 we posed related questions in our surveys when we asked individuals to evaluate the successfulness of their organization's data quality efforts (2011) and effectiveness of their organization's data governance efforts (2008). In both the 2008 and 2011 surveys, the results were similar. On average over half of the responses indicated that results had improved (~57%), about a third of the responses (~33%) specified that results had remained the same, and a small percentage (<10%) of responses were marked as results being unsuccessful or unknown.

This raises an interesting question. Why are perceived data quality ratings not keeping pace with organizations' data quality and governance improvement efforts? One possible explanation is that with each passing year employees are interacting more with information systems and as that interaction increases, they are more sensitive to any data quality issues that impair their decision making capabilities. It may also be that data expectations keep changing due to factors such as increased interest in big data analytics or increased reporting requirements in order to maintain compliance with stricter organizational, financial, and environmental regulations. Another possible explanation is that the volume, velocity, variety and uncertainty surrounding data is increasing faster than the current speed of organizations' data quality and governance efforts. It may also be that poor data quality is a very pervasive problem that will require a sustained effort over many years before we see a substantial increase in the perception of excellence for an organization's data quality.

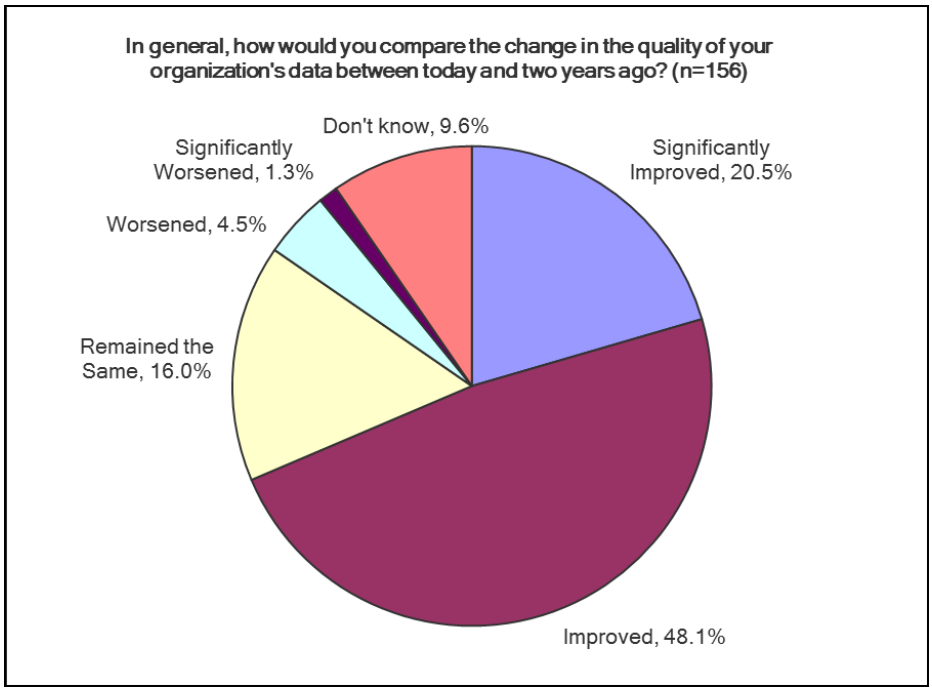


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Has the level of data quality improved over the last two years?

The good news is yes. The majority of survey participants (68.6%) stated data quality in their organization has improved either significantly or very significantly over the last two years. Another 16% reported no change while a small percentage (5.8%) said data quality has worsened or significantly worsened. These results are comparable to the responses we received when we asked a similar question on our 2008 survey indicating that while data quality remains an on-going challenge, organizations are continuing to make progress in addressing issues with their data.



Which data quality activities are the most frequently performed?

Data quality domains were first articulated as part of IQ International's efforts to create an Information Quality Certification. The domains are described as follows.

Information Quality Measurement and Improvement: This domain covers the steps involved in conducting data quality improvement projects. Activities include gathering and analyzing business requirements for data; assessing the quality of data; determining the root causes of data quality issues; developing and implementing information quality improvement plans; preventing and correcting data errors; and implementing information quality controls. In this year's survey as well as in our 2009 survey when we asked a similar question, this domain of activities came in first among our survey participants as their most frequently performed set of activities.

Sustaining Information Quality: This domain focuses on implementing processes and management systems that ensure ongoing information quality. Examples include integrating data quality activities

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into other projects and processes (e.g., data conversion and migration projects, business intelligence efforts, customer data integration activities, enterprise resource planning initiatives, or system development life cycle processes); and continuously monitoring and reporting data quality levels. This domain along with the next two (Information Architecture Quality and Information Quality Value & Business Impact) were ranked second in frequency of performance by our survey participants. In 2009 the Sustaining Information Quality domain also was in the second tier of most frequently performed activities.

Information Architecture Quality: This domain includes the tasks that assure the quality of the data blueprint for an organization. Activities include participating in the establishment of data definitions, standards, and business rules; testing the quality of the information architecture to identify concerns; leading improvement efforts to increase the stability, flexibility, and reuse of the information architecture; and coordinating the management of metadata and reference data. This domain along with the previous (Sustaining Information Quality) ranked second in frequency of performance by our survey participants. In 2009 the Information Architecture Quality domain also was in the second tier of most frequently performed activities.

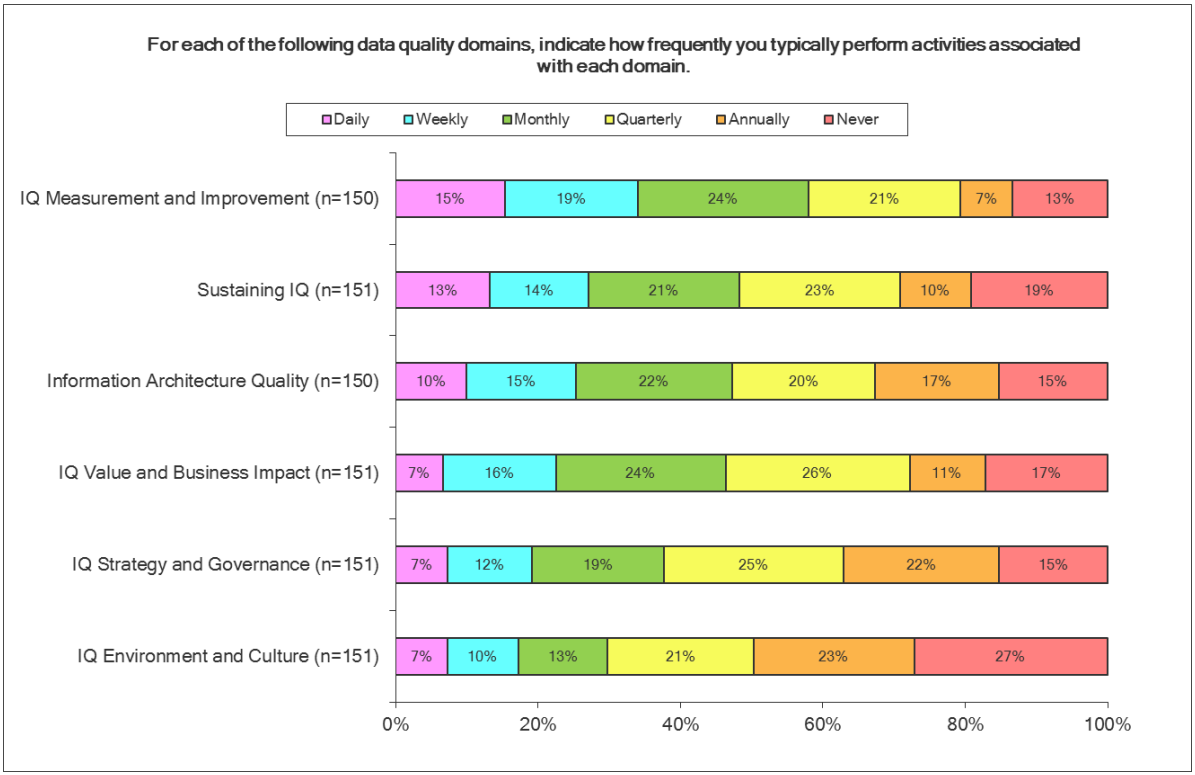
Information Quality Value & Business Impact: This domain consists of the techniques used to determine the effects of data quality on the business as well as the methods for prioritizing information quality projects. Activities include evaluating information quality and business issues; prioritizing information quality initiatives; obtaining decisions on information quality project proposals; and reporting results to demonstrate the value of information quality improvement to the organization. This domain of activities has increased in relative frequency since 2009. Back in 2009 survey participants placed this domain in the third tier of most frequently performed activities. In this year's survey, the responses move this domain up a position to the second tier of most frequently performed activities along with Information Architecture Quality and Sustaining Information Quality.

Information Quality Strategy and Governance: This domain includes the efforts to provide the structures and processes for making decisions about an organization's data as well as ensuring that the appropriate people are engaged to manage information throughout its life cycle. Activities include working with key stakeholders to define and implement information quality principles, policies, and strategies; organizing data governance by naming key roles and responsibilities, establishing decision rights; and building essential relationships with senior leaders in order to improve information quality. In this year's survey as well as in our 2009 survey when we asked a similar question, this domain of activities ranked third among our survey participants as their most frequently performed set of activities.

Information Quality Environment and Culture: This domain provides the background that enables an organization's employees to continuously identify, design, develop, produce, deliver and support

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information quality to meet customer needs. Activities include designing information quality education and training programs; identifying career paths; establishing incentives and controls; promoting information quality as part of business operations; and fostering collaborations across the organization for the purpose of engaging people at all levels in information quality strategies, principles, and practices. In both this year's survey as well as in 2009, this activity domain was cited as the least frequently performed among our survey participants.



It should be noted that although the 2016 relative frequency ranking of the various domains is consistent with what we observed in 2009, there is a definite difference with this year's survey population in terms of the actual stated frequency of performance of these activities. In examining the numbers, this year's respondents reported that they are performing these tasks less frequently on a daily and weekly basis than respondents reported in 2009. In fact, if our 2016 respondents do these activities at all it is more likely to be on a monthly, quarterly or annual basis. The table below summarizes the differences. In addition, a graphic of the 2009 results is available in the Appendix section of this report for readers wishing to see a more detailed comparison. More investigation is needed to determine if these aberrations reflect a difference with this year's survey participants or a real change in who is performing data quality activities along with how often and in what manner these activities are being carried out in organizations. It may be that factors such as automation, maturing data governance operations, and shifting roles and responsibilities are changing how these information quality activities are performed.

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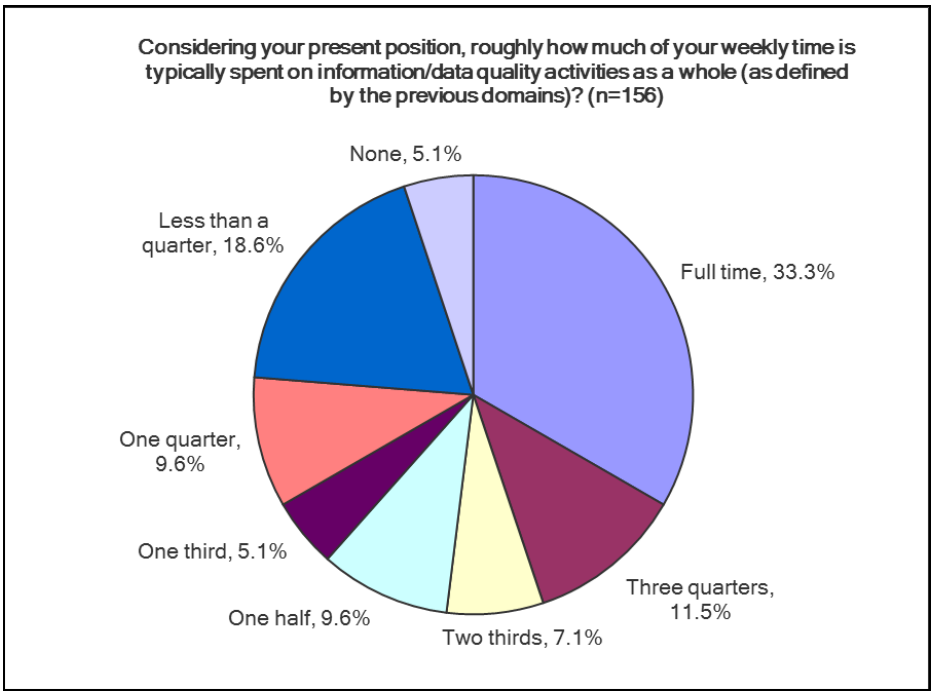
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Frequency	2009 Frequency Range Across the Various Info Quality Domains	2016 Frequency Range Across the Various Info Quality Domains
Daily	~11% - 41%	~7% - 15%
Weekly	~19% - 29%	~10% - 19%
Monthly	~20% - 25%	~13% - 24%
Quarterly	~8% - 15%	~20% - 26%
Annually	~3% - 14%	~7% - 23%
Never	~1% - 19%	~13% - 27%

What percentage of time is spent on data quality activities?

61.5% of our survey participants stated that they spend 50% or more of their time each week on activities associated with these information quality domains. These individuals are the focus of the final section of our report – Compensation and Job Satisfaction of Data Quality Professionals.

It should be noted that this percentage is lower this year as compared to our 2009 survey results. In 2009, 78.6% of respondents reported spending at least 50% or more of their time each week on data quality activities. More investigation is needed to determine whether this difference is due to factors surrounding this year's survey or if it reflects a trend towards fewer individuals spending the majority of their time on data quality efforts.



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Compensation and Satisfaction of Data Quality Professionals

This section of the report discusses various job aspects for data quality professionals. Of the 96 individuals who indicated they spent 50% or more of their weekly time on data quality activities, 93 (~97%) completed questions pertaining to their job titles, compensation, and work satisfaction. Trend charts comparing results from previous surveys are included in the Appendix.

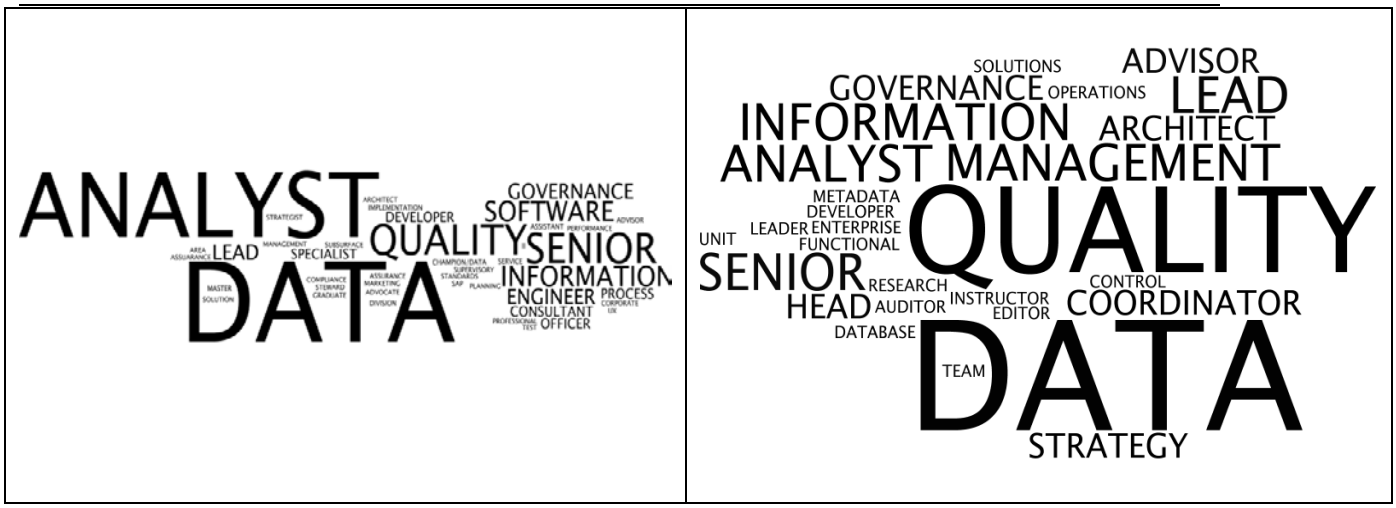
What are the most common job titles for data quality professionals?

Our survey population of data quality professionals provided us with 86 job titles which covered a wide variety of terms. Among Executives/Owners, some common titles were “Chief Data Officer”, Vice President Data Quality Management”, and “Director of Data Governance”. Among Managers, some common titles were “Data (Information) Quality Manager”, “Data Governance Manager”, and “Manager Master Data Management”. The Staff/Individual Contributors and the Team Leaders/Supervisors had the most overlap in their job titles with many of them sharing common titles like “Data Quality Analyst”, “Data (Information) Architect”, and “Data Governance Advisor”.

Executives/Owners (6 entries)	**Managers (29 entries)**
	
Staff/Individual Contributors (33 entries)	**Team Leaders/Supervisors (18 entries)**

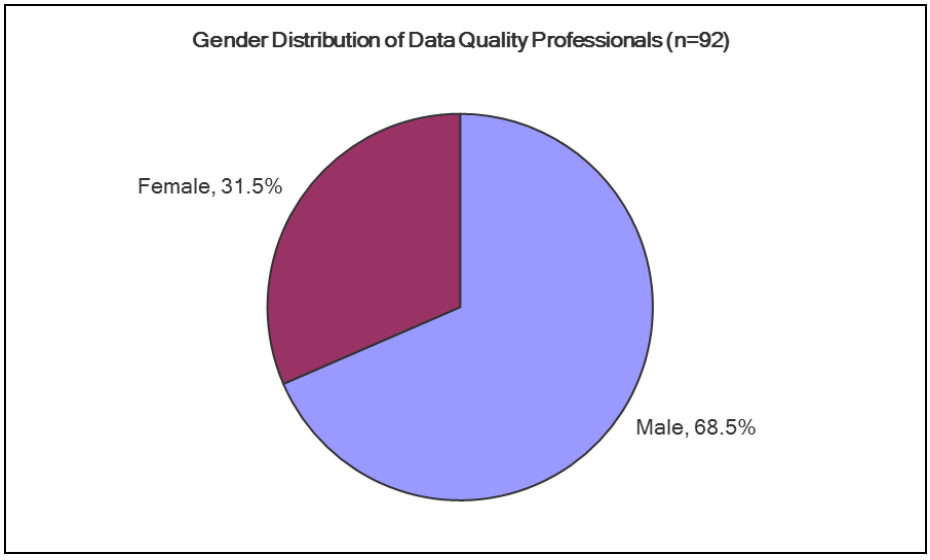
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What is the gender representation among data quality professionals?

Males made up about two thirds of our survey population of individuals who spend 50% or more of their weekly effort on data quality activities. The percentage of women in data quality intensive roles has remained consistently around one-third according to our previous IAIDQ industry reports (i.e., 2009 – 35.9%, 2011 – 30.5%). Our female data quality professional percentage is also in line with other IT gender statistics such as the average percentage of women (30%) working in the tech industry, based on diversity reports published by 11 of the world's largest tech companies in 2015 [http://www.cnet.com/news/women-in-tech-the-numbers-dont-add-up/].



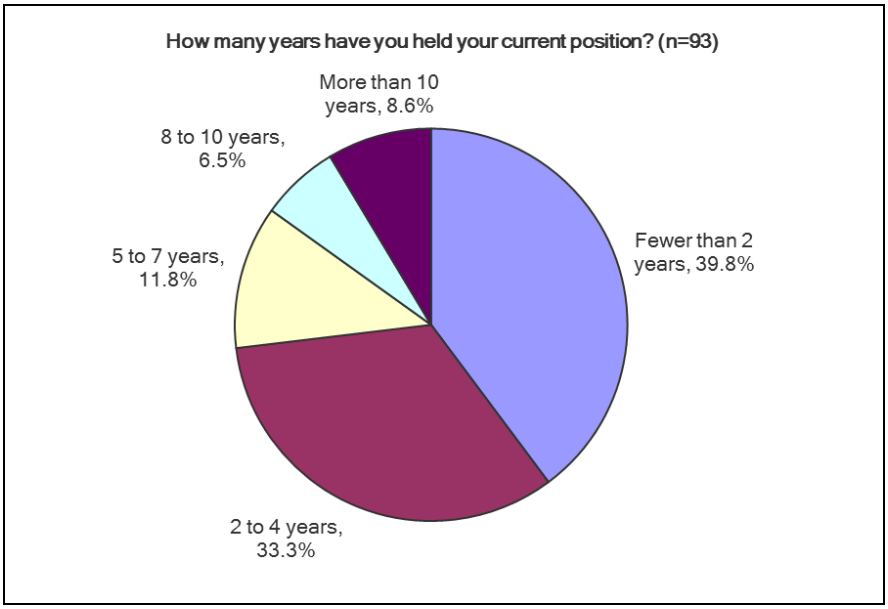
How long have data quality professionals held their positions?

Just under 40% of our survey population of data quality professionals have held their jobs for less than 2 years while a third (33.3%) have held their positions 2 to 4 years, and roughly a quarter (26.9%) had

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held their positions longer than 4 years. The distribution of these responses is comparable to what we observed in our 2009 survey when we asked this question.

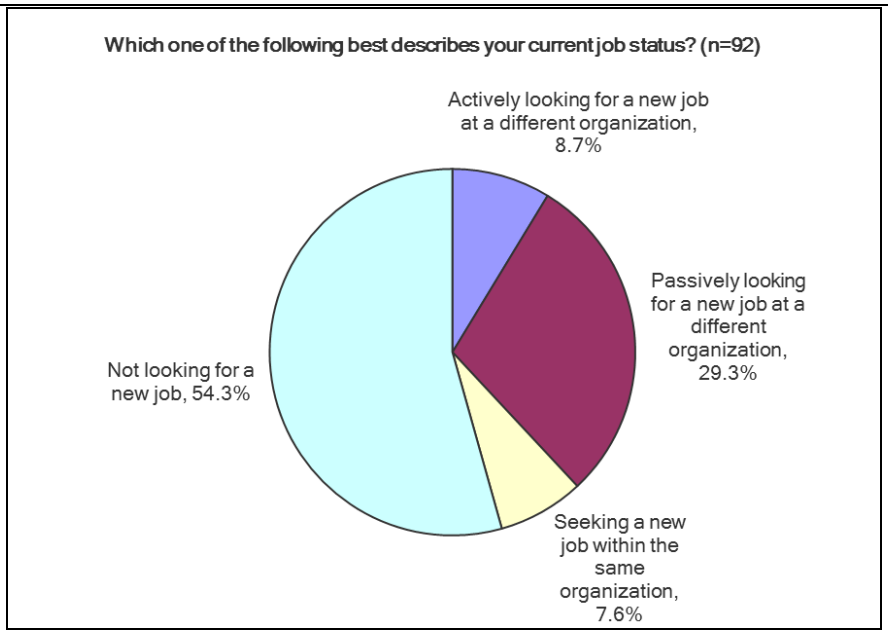


What is the current job status of most data quality professionals?

Over half of our survey population of data quality professionals (54.3%) reported they are planning to stay with their current positions for now. Of the remaining responses, 29.3% are passively looking for a new job at another organization, 8.7% are actively looking for a new job at another organization, and 7.6% would like to stay with their current organization but get a different position. Compared to 2009 when we asked this question, this year's results are similar but there does appear to be a small shift. The percentage of individuals who say they are not looking for a new job decreased slightly (in 2009 it was 61.0%) while the percentage of individuals who say they are passively looking for a new job at a different organization increased by a small amount (in 2009 it was 22.9%).

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What is the average salary of data quality professionals?

We asked respondents to tell us about their compensation. All results are reported in US Dollars using the average currency exchange rate taken on March 15, 2016 from the website: www.poundsterlinglive.com.

2016 Salary (USD) by Organizational Role					
Organizational Role	**N**	**25th percentile**	**Median**	**75th percentile**	**Mean**
Staff / Individual Contributor	24	$57,500	$82,902	$118,875	$86,593
Team Leader / Supervisor	17	$71,986	$79,503	$90,000	$82,087
Manager	23	$91,321	$120,000	$155,000	$118,504
Executive / Owner	7	$165,000	$190,000	$239,000	$190,337
Total (Included for Comparison)	71	$70,558	$90,000	$134,754	$106,080

When we compared these figures to our 2009 survey findings, we saw a pattern we did not expect.

2009 Salary (USD) by Organizational Role					
Organizational Role	**N**	**25th percentile**	**Median**	**75th percentile**	**Mean**
Staff / Individual Contributor	50	$66,731	$92,500	$111,941	$88,716
Team Leader / Supervisor	19	$67,678	$90,000	$104,500	$88,036
Manager	37	$75,047	$100,502	$120,000	$97,792
Executive / Owner	9	$118,089	$150,000	$158,688	$126,293
Total (Included for Comparison)	118	$70,441	$96,500	$120,000	$95,240

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While Managers and Executives/Owners have seen a substantial increase in their salaries over the last seven years, most Staff/Individual Contributors and Team Leaders/Supervisors have seen the salaries for their positions decrease over the last several years. The percentage comparison table shows as one's organizational role rises in rank, the percentage change rises substantially.

Percentage Comparison 2009 to 2016 Salary (USD) by Organizational Role				
Organizational Role	**25th percentile**	**Median**	**75th percentile**	**Mean**
Staff / Individual Contributor	-13.83%	-10.38%	6.19%	-2.39%
Team Leader / Supervisor	6.37%	-11.66%	-13.88%	-6.76%
Manager	21.69%	19.40%	29.17%	21.18%
Executive / Owner	39.73%	26.67%	50.61%	50.71%
Total (Included for Comparison)	0.17%	-6.74%	12.30%	11.38%

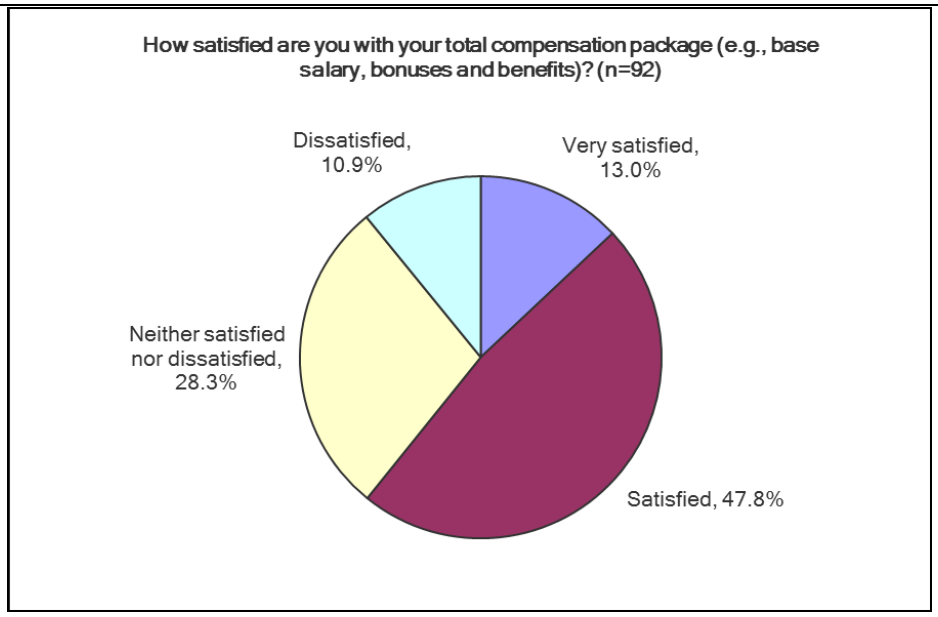
While some of this variation can be due to differences between the 2009 and 2016 survey population, these numbers do suggest that the economy is favoring those at the top more so than those at entry level or mid-level positions.

How satisfied are data quality professionals with their compensation?

The majority of our survey population of data quality professionals (60.8%) indicated they are either satisfied or very satisfied with their compensation with another 28.3% feeling neutral about their compensation, and the remaining 10.9% expressing some level of dissatisfaction with their amount of compensation. These numbers are down from our 2009 survey when 87.5% of our data quality professionals said they were satisfied or very satisfied with their compensation. Given the salary changes noted in the previous question, this may point to growing dissatisfaction with how compensation is distributed among the different levels of data quality professionals.

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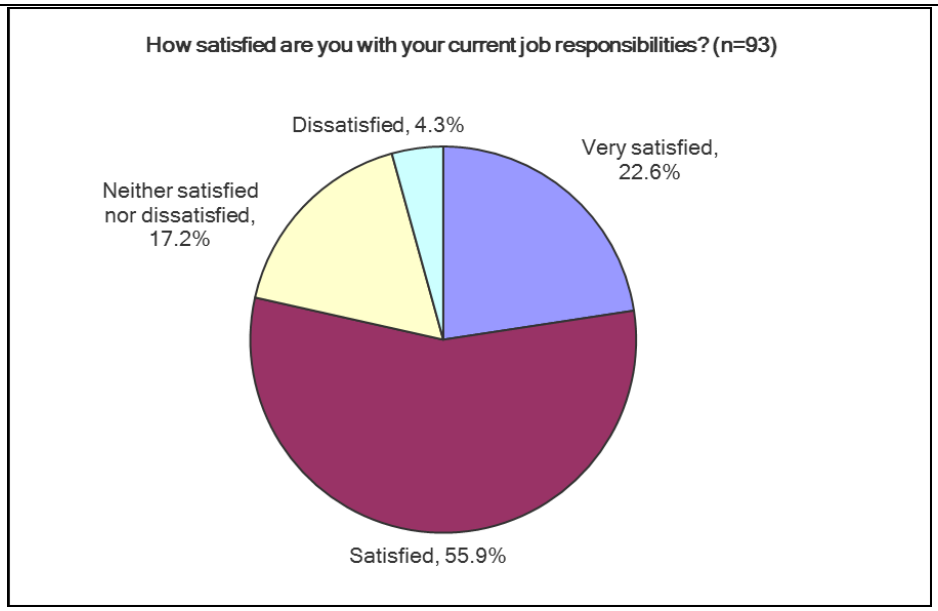


How satisfied are data quality professionals with their job responsibilities?

The majority of our survey population of data quality professionals (78.5%) stated they are either satisfied or very satisfied with their job responsibilities with another 17.2% feeling neutral about their job responsibilities, and the remaining 4.3% expressing dissatisfaction with their present set of job responsibilities. Compared to 2009 when we asked this question, the results are fairly similar (the response percentages in 2009 were 73.1% either satisfied or very satisfied, 25.2% somewhat dissatisfied, and 1.7% very dissatisfied).

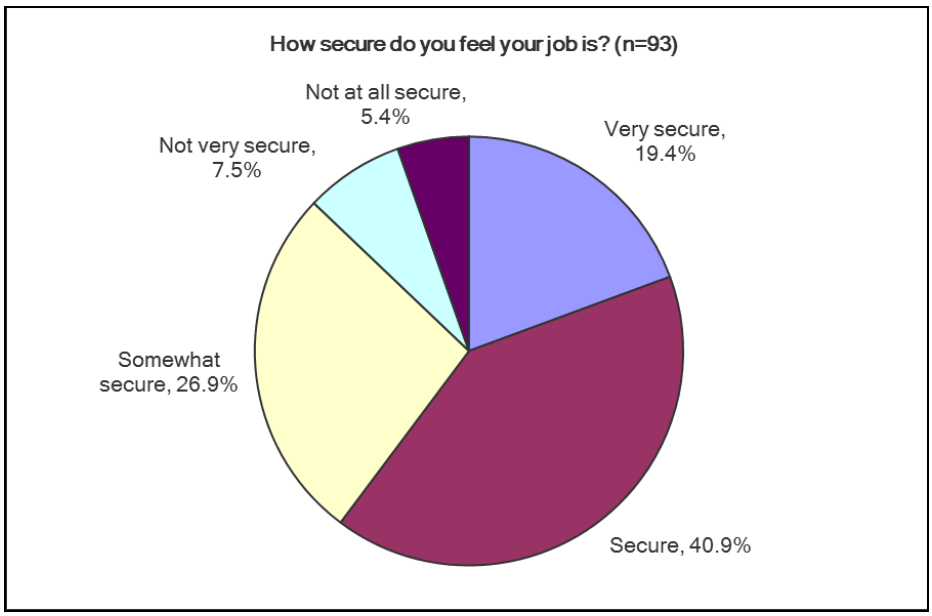
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How secure do data quality professionals feel about their jobs?

The majority of our survey population of data quality professionals (60.3%) reported they are either secure or very secure regarding their present position with another 26.9% feeling somewhat secure, and the remaining 12.9% expressing serious concerns about their job security. In 2009 when we asked a similar question, nearly 78% of data quality professionals said they felt either secure or very secure about their current position. It appears that there has been a shift between 2009 and 2016 with a larger percentage of today's data quality professionals feeling less secure about their positions.

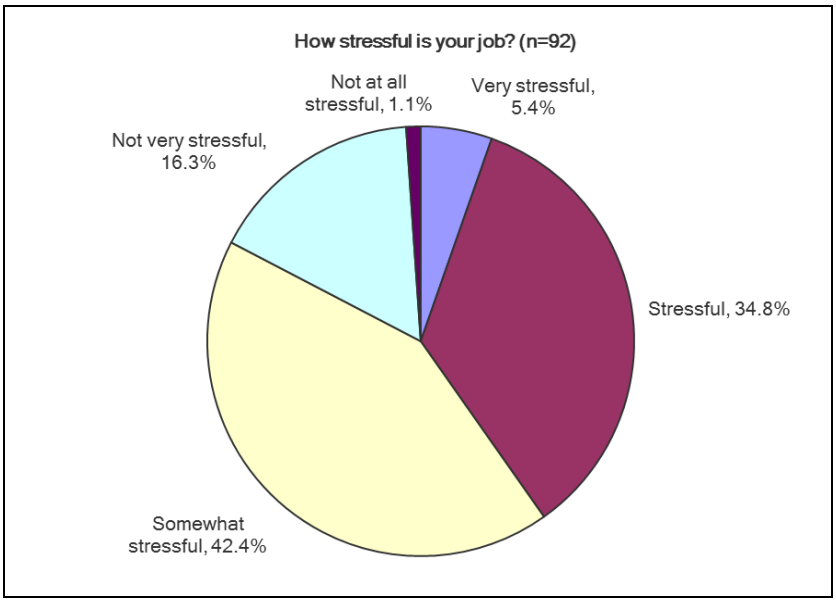


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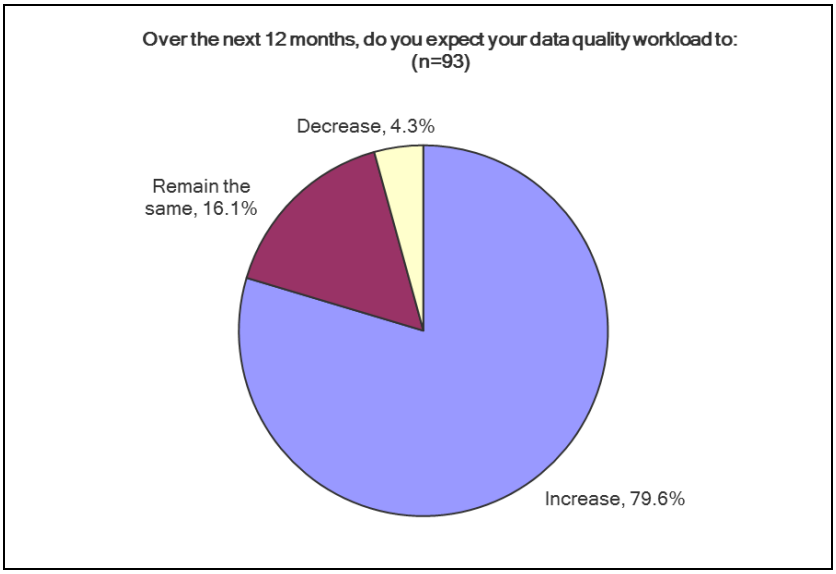
How much stress do data quality professionals feel about their jobs?

Stress seems to be a part of many professional career paths and the data quality profession is no exception. Over 82% of our survey population of data quality professionals indicated experiencing some level of significant stress regarding their jobs. Only 17.4% of our survey population of data quality professionals said they experience little to no stress in their work activities.



How is the workload of data quality professionals changing?

Part of the reason for the stress may be due to the anticipated changes in workload. Nearly 80% of our survey population of data quality professionals expected that their data quality activities will increase over the next 12 months with another 16.1% predicting their work load will remain the same. Only 4.3% thought their data quality workload would decrease next year.



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What are the biggest challenges faced by data quality professionals?

The number one obstacle faced by our survey population of data quality professionals (73.9%) is the challenge posed by how to demonstrate the value of data quality to their employers. Other commonly chosen obstacles include "finding/hiring qualified data quality professionals" (42.4%), "having a clear career/advancement path" (43.1%), and "keeping one's professional and technical skills up-to-date" (39.1%). Several respondents had other concerns that they wished to share. These additional challenges included:

- Hiring qualified data quality professionals with cross discipline knowledge such as healthcare.
- Keeping IT's role to a minimum.
- Lack of funding for appropriate tools.
- Lack of rewards for innovation.
- Lack of people with a genuine passion for data quality.
- Lack of understanding of the complexities of data quality management such as regulatory risks, executive vision and planning, need for sustainable and repeatable data quality processes, and commitment to data stewardship as a profession.



Compared to our 2009 survey, while several of these obstacles maintain a similar ranking, there are some differences.

1. Today's data quality professionals are less worried about the economy. In 2009 the economy ranked as #5 on the list of biggest challenges. In 2016 the economy came in as #9.

2. However, despite the improved outlook on the economy, today's data quality professionals are more worried about their compensation. In 2009 compensation ranked as #9 on the list of biggest

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challenges. In 2016 compensation came in as #6. This may indicate that while the economy as a whole is doing better, individuals and their compensation may not be keeping up.

3. Today's data quality professionals are less worried about the availability of relevant data quality (DQ) training and education. In 2009 concerns about the availability of relevant DQ training ranked as #2 on the list of biggest challenges. In 2016 "availability of relevant DQ training" came in as #5. Online resources, the Information Quality Certified Professional Credential, and web based programs like UALR's Graduate Program in Information Quality may all be contributing factors to this change.

4. However, despite the improved outlook on the availability of relevant DQ training and education, today's data quality professionals are more concerned about keeping their skills current. In 2009 concerns over keeping skills up-to-date ranked #6 on the list of biggest challenges. In 2016 "keeping skills up-to-date" came in as #4.

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Closing Remarks and Recommendations: The Agenda Forward

Most participants in this survey rate their organizations' quality of data as either fair or good. Fewer than 10% of them assess their organization's data as excellent. Moreover, for many organizations, several of the data governance objectives have only been modestly achieved. Several important goals such as "Reduce costs and increase effectiveness through coordination of efforts", "Establish clear decision rules and decision-making processes for shared data", "Ensure transparency of processes for the creation and enforcement of data policies", and "Train management and staff in best practices for working with data" received "Poor" or "Very Poor" accomplishment scores from 30% or more of this year's respondents.

So what can be done to increase the number of organizations that attain excellence in their data quality and reach more of their data governance goals? Several recommendations can be made but they all require the commitment of stakeholders such as universities, professional groups like IQ International, and the organizations themselves in order to carry them out:

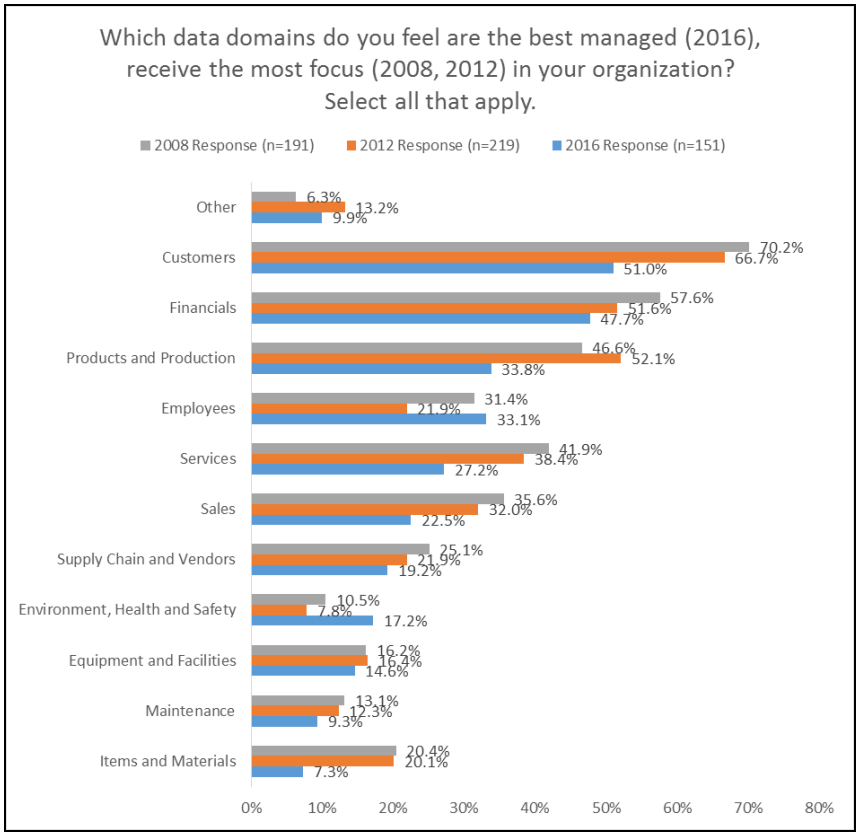
1. More sharing of best practices is needed, particularly when it comes to demonstrating the value of data quality to employers. More work is needed to develop a repository of anecdotes and case studies demonstrating the gains from better data quality, to help persuade corporate leaders.
2. More open education opportunities are needed to make it easier for individuals at all organizational levels to acquire data quality knowledge and skills. Education portals such as Coursera.org, edX.org, BigDataUniversity.com, and ASQ Knowledge Center are good examples of training materials (e.g., tools, publications, standards, case studies, webcasts, videos, and research) provided on the web for worldwide access.
3. Achieving and sustaining high quality data requires constancy of purpose and engagement at all organizational levels. Organizations need to expand their efforts to build a strong data quality environment and culture, while continuing to shift their data quality focus to the revenue side.
4. Better career management is needed. More work is needed to understand how best to organize and staff data governance and data quality initiatives, along with more information on how to reward and recognize data quality and governance practitioners and professionals.
5. More research is needed using alternative approaches to the traditional survey approach. Getting good participation was one of the difficulties with this year's study. It is understandable. As people become busier and as more organizations use web/e-mail to send out surveys to gauge service and product satisfaction, the more survey-weary people become. The result is that our discipline needs to investigate other ways to gather evidence for industry studies such as this one.

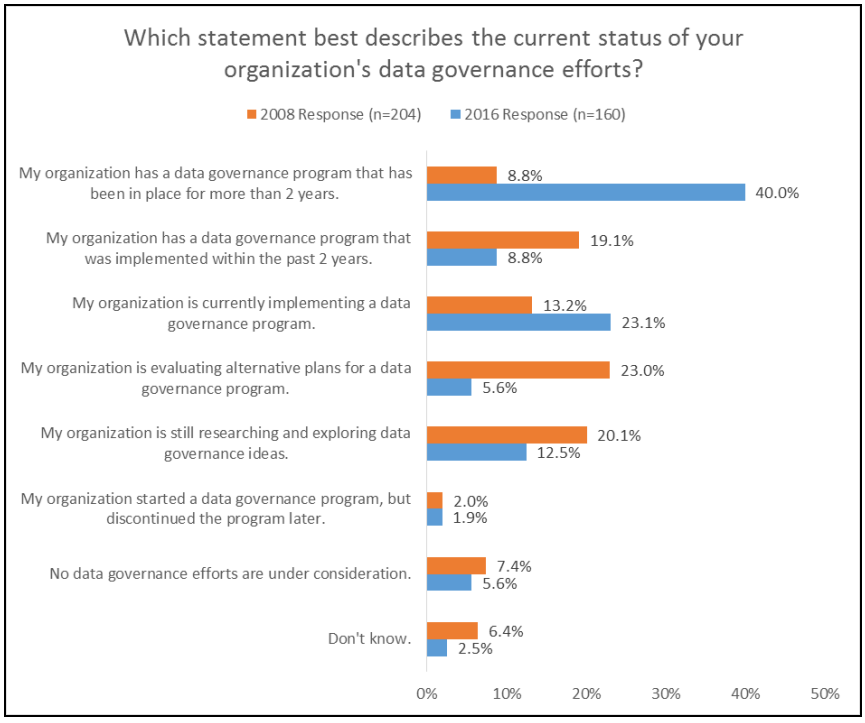
In closing, we hope that this report will be useful in generating more discussion, ideas and action, to help organizations advance their data governance efforts, improve the quality of their data, and provide rewarding career paths for the current and next generation of individuals engaged in these efforts.

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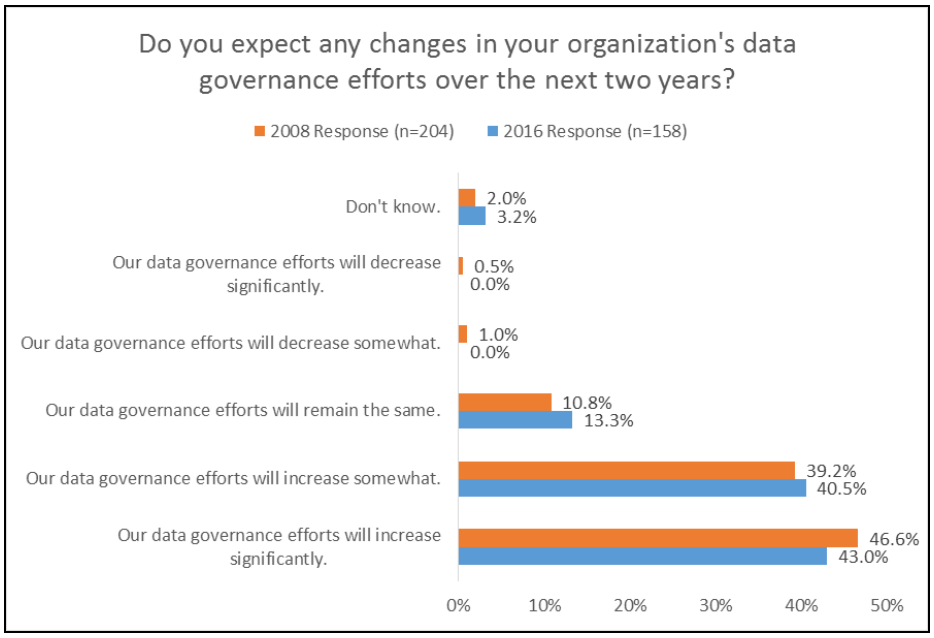
Appendix – Additional Trend Comparison Charts

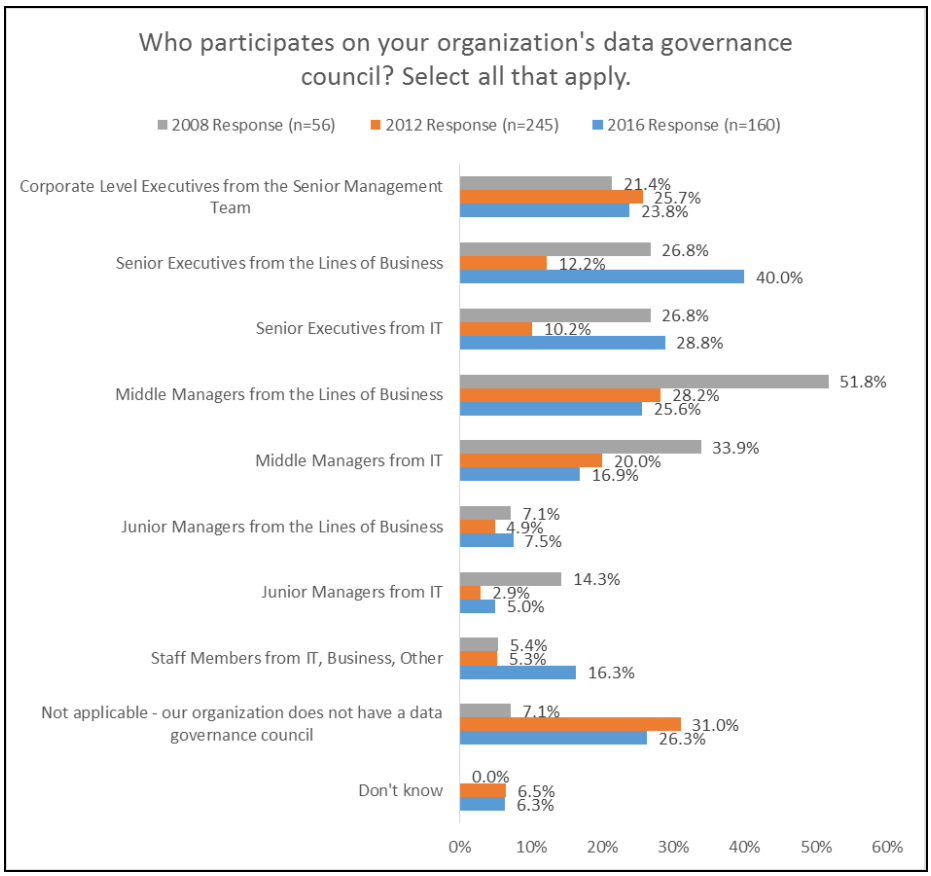




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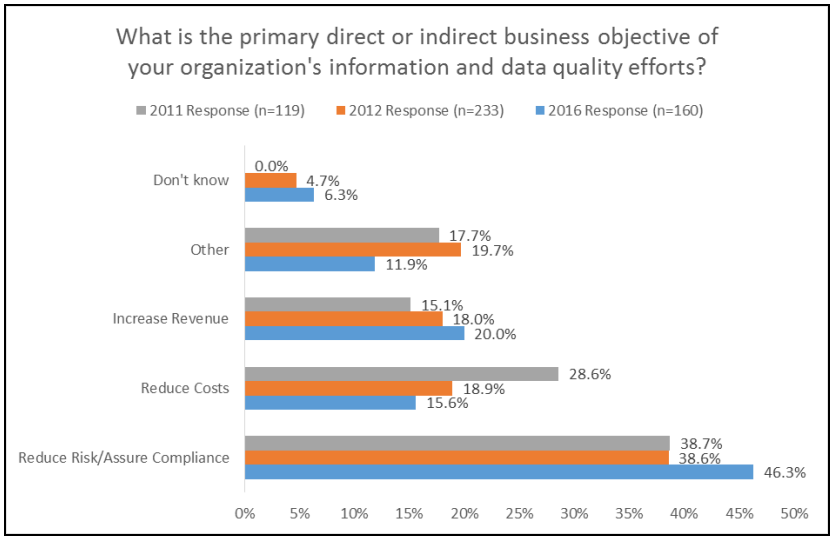
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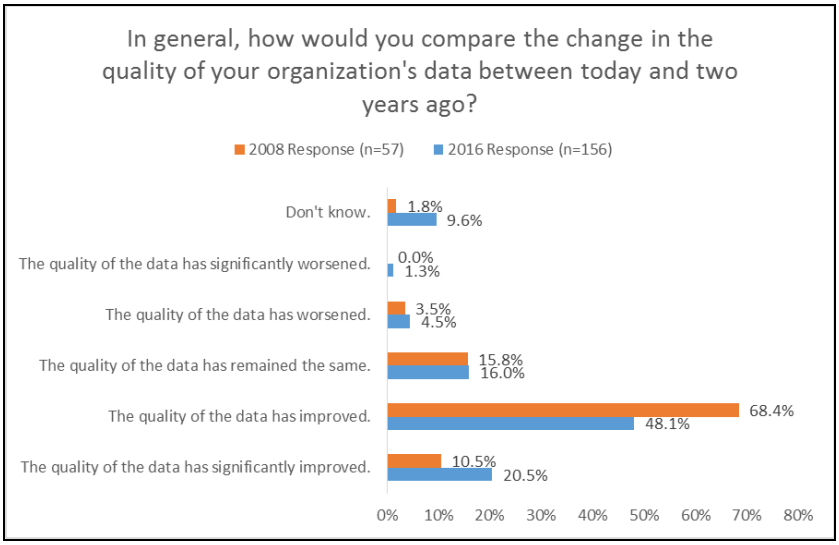


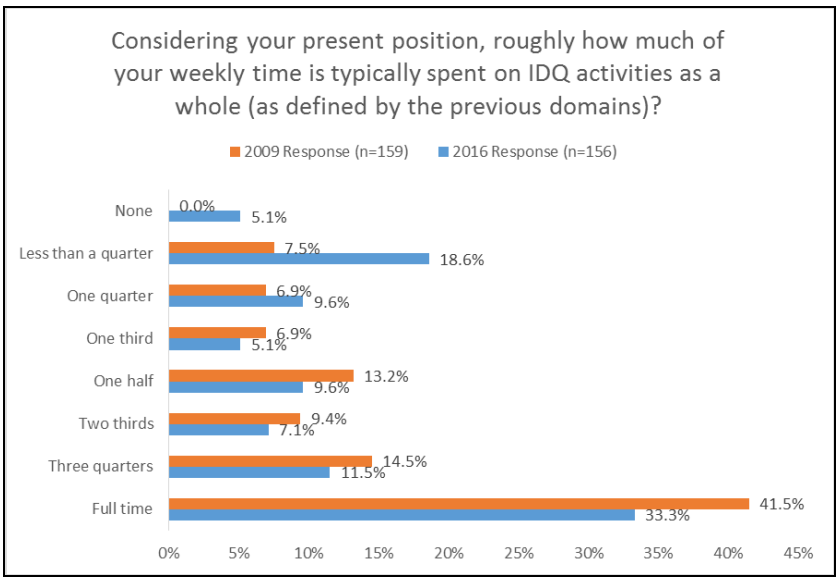


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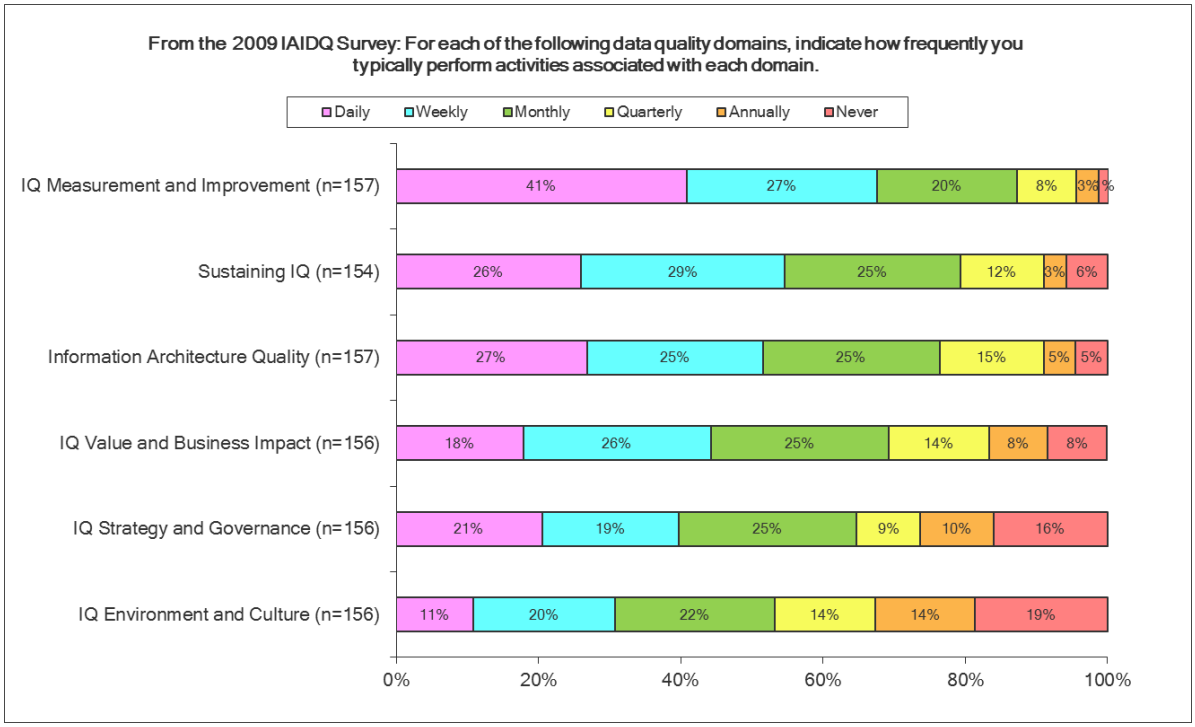


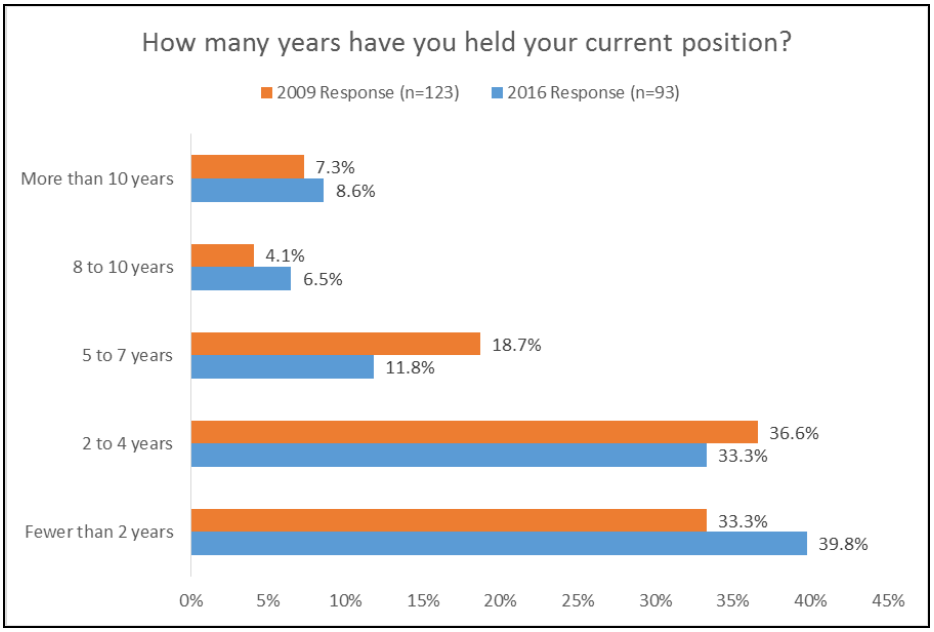




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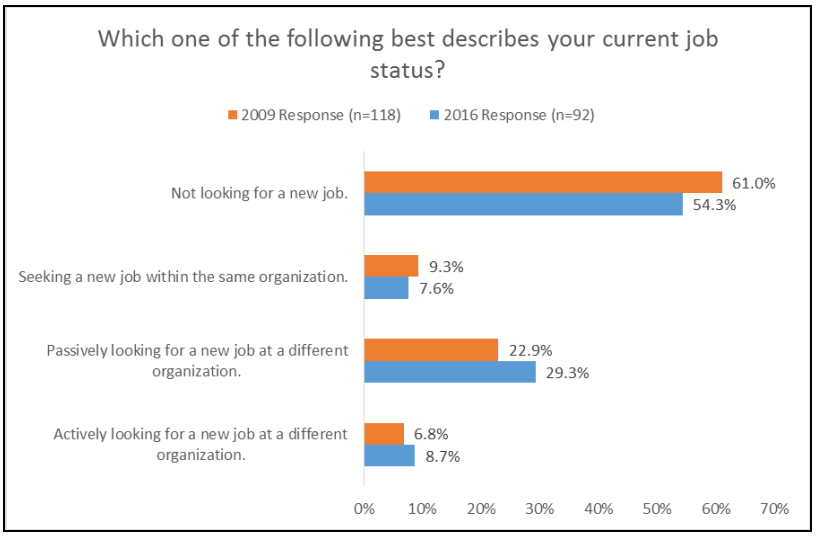
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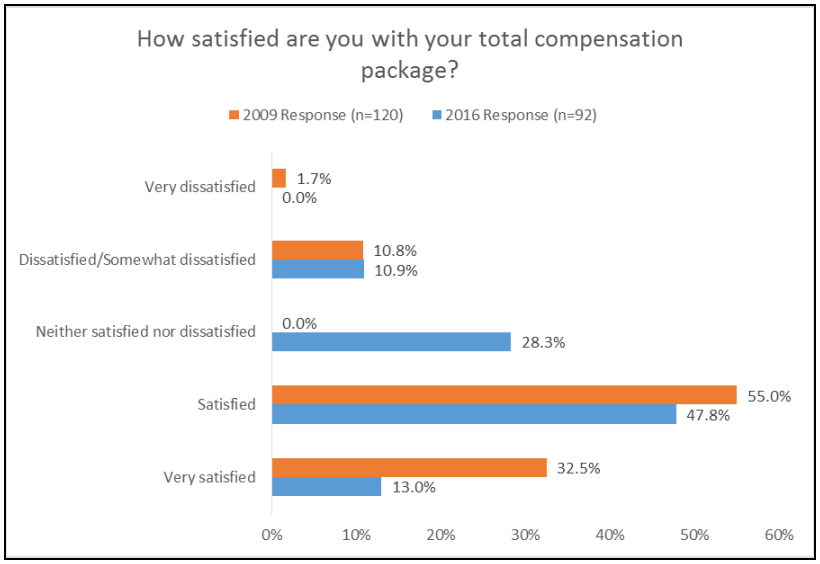


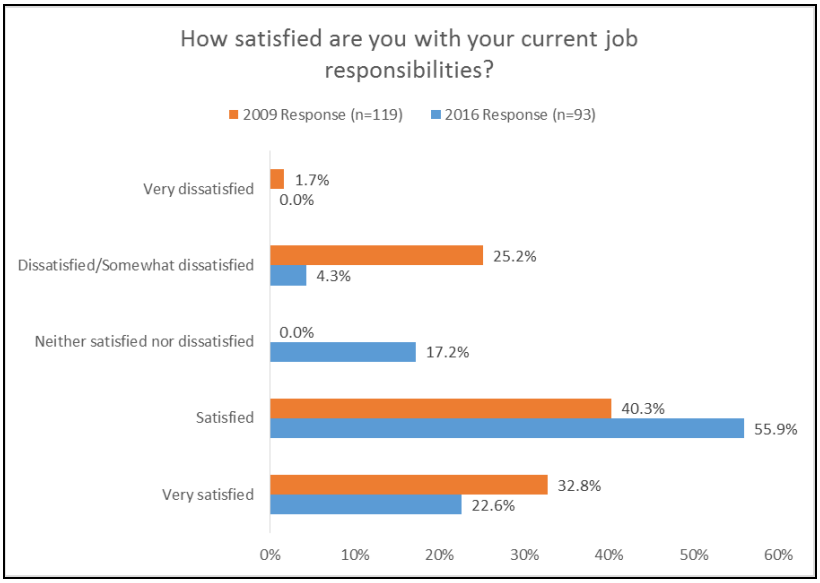


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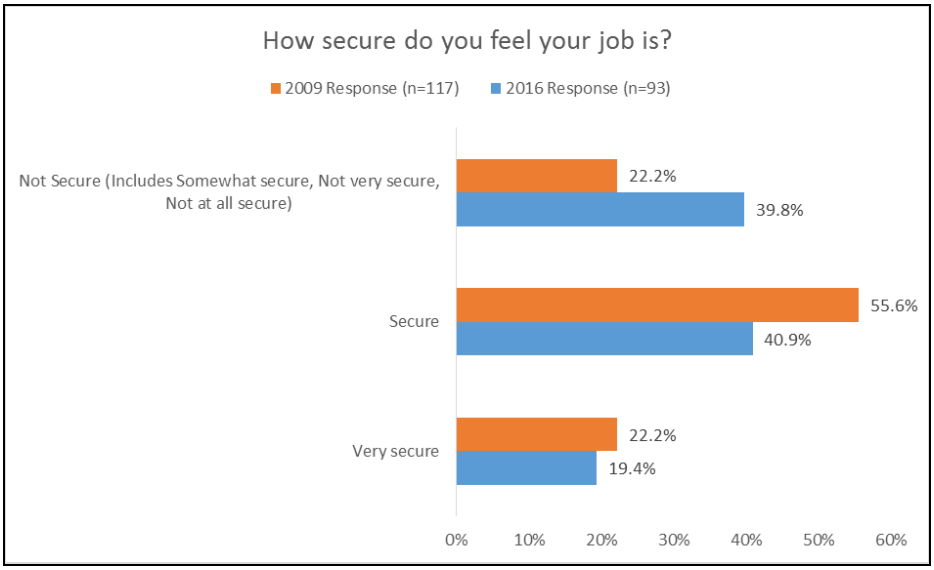


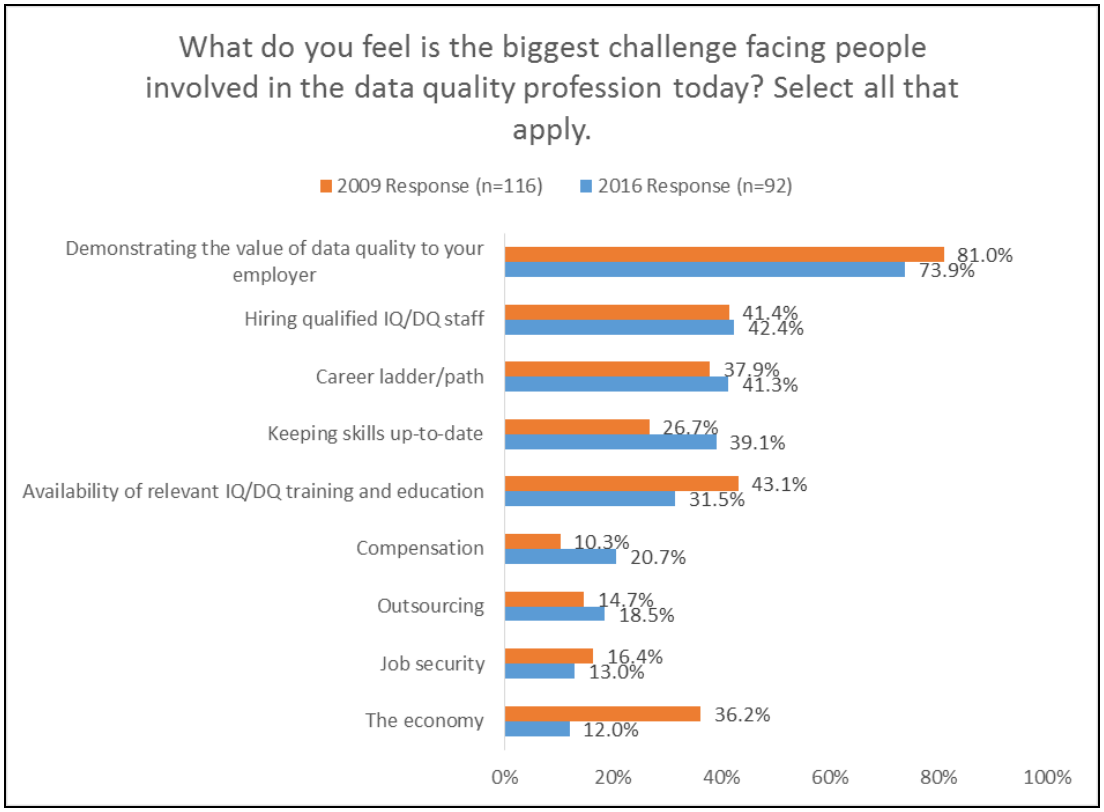




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